

ELECTRIC CIRCUITS V - REFERENCES



Tony R. Kuphaldt
Bellingham Technical College

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1.7: Time Constant Equations

Value of time constant in series RC and RL circuits

Time constant in seconds = RC

Time constant in seconds = L/R

Calculating voltage or current at specified time

Universal Time Constant Formula

$$\text{Change} = (\text{Final} - \text{Start}) \left(1 - \frac{1}{e^{t/\tau}} \right)$$

Where,

Final = Value of calculated variable after infinite time
(its *ultimate* value)

Start = Initial value of calculated variable

e = Euler's number (≈ 2.7182818)

t = Time in seconds

τ = Time constant for circuit in seconds

Calculating time at specified voltage or current

$$t = -\tau \left(\ln \left(1 - \frac{\text{Change}}{\text{Final} - \text{Start}} \right) \right)$$

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1.8: AC Circuit Equations

Inductive reactance

$$X_L = 2\pi fL$$

Where,

X_L = Inductive reactance in ohms

f = Frequency in hertz

L = Inductance in henrys

Capacitive reactance

$$X_C = \frac{1}{2\pi fC}$$

Where,

X_C = Inductive reactance in ohms

f = Frequency in hertz

C = Capacitance in farads

Impedance in relation to R and X

$$Z_L = R + jX_L$$

$$Z_C = R - jX_C$$

Ohm's Law for AC

$$E = IZ \quad I = \frac{E}{Z} \quad Z = \frac{E}{I}$$

Where,

E = Voltage in volts

I = Current in amperes (amps)

Z = Impedance in ohms

Series and Parallel Impedances

$$Z_{\text{series}} = Z_1 + Z_2 + \dots + Z_n$$

$$Z_{\text{parallel}} = \frac{1}{\frac{1}{Z_1} + \frac{1}{Z_2} + \dots + \frac{1}{Z_n}}$$

NOTE: All impedances must be calculated in *complex* number form for these equations to work.

Resonance

$$f_{\text{resonant}} = \frac{1}{2\pi \sqrt{LC}}$$

NOTE: This equation applies to a non-resistive LC circuit. In circuits containing resistance as well as inductance and capacitance, this equation applies only to series configurations and to parallel configurations where R is very small.

AC power

$$\mathbf{P = true\ power} \quad \mathbf{P = I^2R} \quad \mathbf{P = \frac{E^2}{R}}$$

*Measured in units of **Watts***

$$\mathbf{Q = reactive\ power} \quad \mathbf{Q = I^2X} \quad \mathbf{Q = \frac{E^2}{X}}$$

*Measured in units of **Volt-Amps-Reactive (VAR)***

$$\mathbf{S = apparent\ power} \quad \mathbf{S = I^2Z} \quad \mathbf{S = \frac{E^2}{Z}} \quad \mathbf{S = IE}$$

*Measured in units of **Volt-Amps (VA)***

$$\mathbf{P = (IE)(power\ factor)}$$

$$\mathbf{S = \sqrt{P^2 + Q^2}}$$

$$\mathbf{Power\ factor = \cos\ (Z\ phase\ angle)}$$

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4.2: Arithmetic Properties

The associative property

In addition and multiplication, terms may be arbitrarily *associated* with each other through the use of parentheses:

$$a + (b + c) = (a + b) + c \qquad a(bc) = (ab)c$$

The commutative property

In addition and multiplication, terms may be arbitrarily interchanged, or *commutated*:

$$a + b = b + a \qquad ab = ba$$

The distributive property

$$a(b + c) = ab + ac$$

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4.3: Properties of Exponents

$$a^m a^n = a^{m+n}$$

$$(ab)^m = a^m b^m$$

$$(a^m)^n = a^{mn}$$

$$\frac{a^m}{a^n} = a^{m-n}$$

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4.4: Radicals

Definition of a radical

When people talk of a “square root,” they’re referring to a radical with a root of 2. This is mathematically equivalent to a number raised to the power of $1/2$. This equivalence is useful to know when using a calculator to determine a strange root. Suppose for example you needed to find the fourth root of a number, but your calculator lacks a “4th root” button or function. If it has a y^x function (which any scientific calculator should have), you can find the fourth root by raising that number to the $1/4$ power, or $x^{0.25}$.

$$\sqrt[x]{a} = a^{1/x}$$

It is important to remember that when solving for an *even* root (square root, fourth root, etc.) of any number, there are *two* valid answers. For example, most people know that the square root of nine is three, but *negative* three is also a valid answer, since $(-3)^2 = 9$ just as $3^2 = 9$.

Properties of radicals

$$\sqrt[x]{a^x} = a \qquad \sqrt[x]{a^x} = a$$

$$\sqrt[x]{ab} = \sqrt[x]{a} \sqrt[x]{b}$$

$$\sqrt[x]{\frac{a}{b}} = \frac{\sqrt[x]{a}}{\sqrt[x]{b}}$$

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4.5: Important Constants

Euler's number

Euler's constant is an important value for exponential functions, especially scientific applications involving decay (such as the decay of a radioactive substance). It is especially important in calculus due to its uniquely self-similar properties of integration and differentiation.

e approximately equals:
2.71828 18284 59045 23536 02874 71352 66249 77572 47093 69996

$$e = \sum_{k=0}^{\infty} \frac{1}{k!}$$

$$\frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots \frac{1}{n!}$$

Pi

Pi (π) is defined as the ratio of a circle's circumference to its diameter.

Pi approximately equals:
3.14159 26535 89793 23846 26433 83279 50288 41971 69399 37511

Note: For both Euler's constant (e) and pi (π), the spaces shown between each set of five digits have no mathematical significance. They are placed there just to make it easier for your eyes to "piece" the number into five-digit groups when manually copying.

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4.8: The Quadratic Formula

For a polynomial expression in
the form of: $ax^2 + bx + c = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

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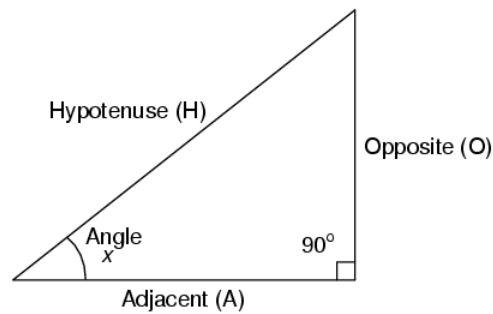
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5.1: Right Triangle Trigonometry



A *right triangle* is defined as having one angle precisely equal to 90° (a *right angle*).

Trigonometric Identities

$$\begin{array}{llll} \sin x = \frac{O}{H} & \cos x = \frac{A}{H} & \tan x = \frac{O}{A} & \tan x = \frac{\sin x}{\cos x} \\ \csc x = \frac{H}{O} & \sec x = \frac{H}{A} & \cot x = \frac{A}{O} & \cot x = \frac{\cos x}{\sin x} \end{array}$$

H is the *Hypotenuse*, always being opposite the right angle. Relative to angle x , O is the *Opposite* and A is the *Adjacent*.

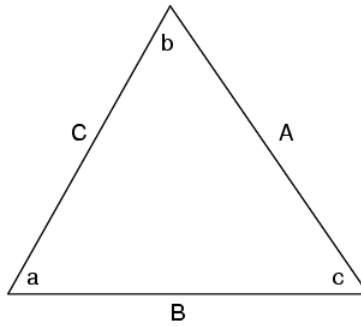
“Arc” functions such as “arcsin”, “arccos”, and “arctan” are the complements of normal trigonometric functions. These functions return an angle for a ratio input. For example, if the tangent of 45° is equal to 1, then the “arctangent” (arctan) of 1 is 45° . “Arc” functions are useful for finding angles in a right triangle if the side lengths are known.

The Pythagorean Theorem

$$H^2 = A^2 + O^2$$

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5.2: Non-right Triangle Trigonometry



The Law of Sines (for any triangle)

$$\frac{\sin a}{A} = \frac{\sin b}{B} = \frac{\sin c}{C}$$

The Law of Cosines (for any triangle)

$$A^2 = B^2 + C^2 - (2BC)(\cos a)$$

$$B^2 = A^2 + C^2 - (2AC)(\cos b)$$

$$C^2 = A^2 + B^2 - (2AB)(\cos c)$$

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6.9: Antiderivatives of Power Functions of e

$$\int e^x dx = e^x + c$$

Note: this is a very unique and useful property of e. As in the case of derivatives, the antiderivative of such a function is that same function. In the case of the antiderivative, a constant term “c” is added to the end as well.

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7.7: SPICE Quirks

“Garbage in, garbage out.”

—Anonymous

SPICE is a very reliable piece of software, but it does have its little quirks that take some getting used to. By “quirk” I mean a demand placed upon the user to write the source file in a particular way in order for it to work without giving error messages. I do *not* mean any kind of fault with SPICE which would produce erroneous or misleading results: that would be more properly referred to as a “bug.” Speaking of bugs, SPICE has a few of them as well.

Some (or all) of these quirks may be unique to SPICE version 2g6, which is the only version I’ve used extensively. They may have been fixed in later versions.

A good beginning

SPICE demands that the source file begins with something other than the first “card” in the circuit description “deck.” This first character in the source file can be a linefeed, title line, or a comment: there just has to be *something* there before the first component-specifying line of the file. If not, SPICE will refuse to do an analysis at all, claiming that there is a serious error (such as improper node connections) in the “deck.”

A good ending

SPICE demands that the `.end` line at the end of the source file not be terminated with a linefeed or carriage return character. In other words, when you finish typing “`.end`” you should not hit the **[Enter]** key on your keyboard. The cursor on your text editor should stop immediately to the right of the “d” after the “`.end`” and go no further. Failure to heed this quirk will result in a “*missing .end card*” error message at the end of the analysis output. The actual circuit analysis is not affected by this error, so I normally ignore the message. However, if you’re looking to receive a “perfect” output, you must pay heed to this idiosyncrasy.

Must have a node 0

You are given much freedom in numbering circuit nodes, but you *must* have a node 0 somewhere in your netlist in order for SPICE to work. Node 0 is the default node for circuit ground, and it is the point of reference for all voltages specified at single node locations.

When simple DC analysis is performed by SPICE, the output will contain a listing of voltages at all non-zero nodes in the circuit. The point of reference (ground) for all these voltage readings is node 0. For example:

```
node voltage node voltage ( 1) 15.0000 ( 2) 0.6522
```

In this analysis, there is a DC voltage of 15 volts between node 1 and ground (node 0), and a DC voltage of 0.6522 volts between node 2 and ground (node 0). In both these cases, the voltage polarity is negative at node 0 with reference to the other node (in other words, both nodes 1 and 2 are positive with respect to node 0).

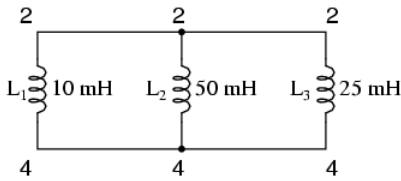
Avoid open circuits

SPICE cannot handle open circuits of any kind. If your netlist specifies a circuit with an open voltage source, for example, SPICE will refuse to perform an analysis. A prime example of this type of error is found when “connecting” a voltage source to the input of a voltage-dependent source (used to simulate an operational amplifier). SPICE needs to see a complete path for current, so I usually tie a high-value resistor (call it *rbogus*!) across the voltage source to act as a minimal load.

Avoid certain component loops

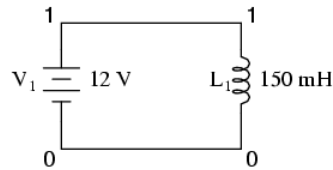
SPICE cannot handle certain uninterrupted loops of components in a circuit, namely voltage sources and inductors. The following loops will cause SPICE to abort analysis:

Parallel inductors



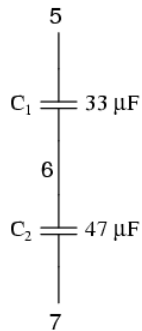
```
netlist l1 2 4 10m l2 2 4 50m l3 2 4 25m
```

Voltage source / inductor loop



```
netlist v1 1 0 dc 12 l1 1 0 150m
```

Series capacitors

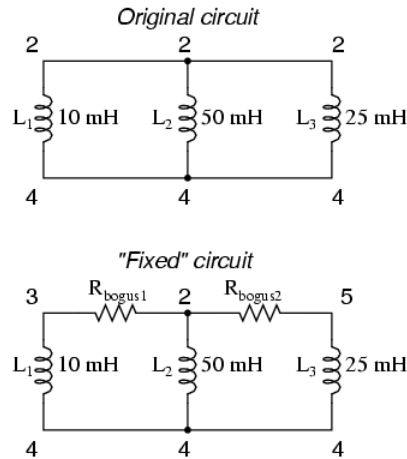


```
netlist c1 5 6 33u c2 6 7 47u
```

The reason SPICE can't handle these conditions stems from the way it performs DC analysis: by treating all inductors as shorts and all capacitors as opens. Since short-circuits (0Ω) and open circuits (infinite resistance) either contain or generate mathematical infinities, a computer simply cannot deal with them, and so SPICE will discontinue analysis if any of these conditions occur.

In order to make these component configurations acceptable to SPICE, you must insert resistors of appropriate values into the appropriate places, eliminating the respective short-circuits and open-circuits. If a series resistor is required, choose a very low resistance value. Conversely, if a parallel resistor is required, choose a very high resistance value. For example:

To fix the parallel inductor problem, insert a very low-value resistor in series with each offending inductor.

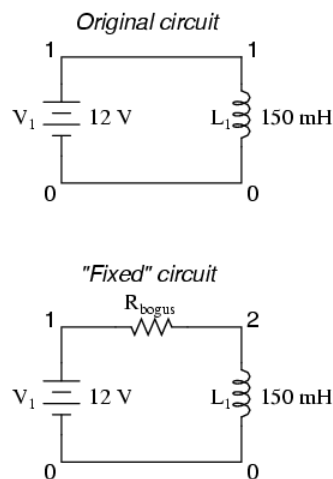


```
original netlist l1 2 4 10m l2 2 4 50m l3 2 4 25m
```

```
fixed netlist rbogus1 2 3 1e-12 rbogus2 2 5 1e-12 l1 3 4 10m l2 2 4 50m l3 5 4 25m
```

The extremely low-resistance resistors R_{bogus1} and R_{bogus2} (each one with a mere 1 pico-ohm of resistance) “break up” the direct parallel connections that existed between L_1 , L_2 , and L_3 . It is important to choose very low resistances here so that circuit operation is not substantially impacted by the “fix.”

To fix the voltage source / inductor loop, insert a very low-value resistor in series with the two components.

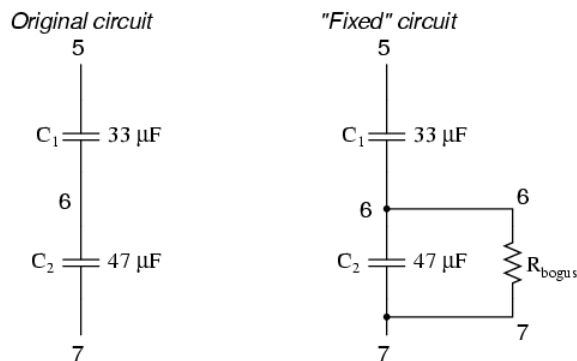


```
original netlist v1 1 0 dc 12 l1 1 0 150m
```

```
fixed netlist v1 1 0 dc 12 l1 2 0 150m rbogus 1 2 1e-12
```

As in the previous example with parallel inductors, it is important to make the correction resistor (R_{bogus}) very low in resistance, so as to not substantially impact circuit operation.

To fix the series capacitor circuit, one of the capacitors must have a resistor shunting across it. SPICE requires a DC current path to each capacitor for analysis.



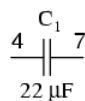
```
original netlist c1 5 6 33u c2 6 7 47u
```

```
fixed netlist c1 5 6 33u c2 6 7 47u rbogus 6 7 9e12
```

The R_{bogus} value of 9 Tera-ohms provides a DC current path to C_1 (and around C_2) without substantially impacting the circuit's operation.

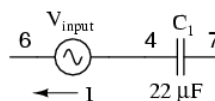
Current measurement

Although printing or plotting of voltage is quite easy in SPICE, the output of current values is a bit more difficult. Voltage measurements are specified by declaring the appropriate circuit nodes. For example, if we desire to know the voltage across a capacitor whose leads connect between nodes 4 and 7, we might make our `.print` statement look like this:



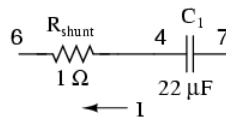
```
c1 4 7 22u .print ac v(4,7)
```

However, if we wanted to have SPICE measure the *current* through that capacitor, it wouldn't be quite so easy. Currents in SPICE must be specified in relation to a voltage source, not any arbitrary component. For example:



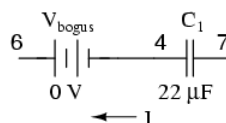
```
c1 4 7 22u vinput 6 4 ac 1 sin .print ac i(vinput)
```

This `.print` card instructs SPICE to print the current through voltage source V_{input} , which happens to be the same as the current through our capacitor between nodes 4 and 7. But what if there is no such voltage source in our circuit to reference for current measurement? One solution is to insert a shunt resistor into the circuit and measure voltage across it. In this case, I have chosen a shunt resistance value of $1\ \Omega$ to produce 1 volt per amp of current through C_1 :



```
c1 4 7 22u rshunt 6 4 1 .print ac v(6,4)
```

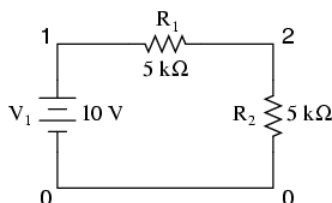
However, the insertion of an extra resistance into our circuit large enough to drop a meaningful voltage for the intended range of current might adversely affect things. A better solution for SPICE is this, although one would never seek such a current measurement solution in real life:



```
c1 4 7 22u vbogus 6 4 dc 0 .print ac i(vbogus)
```

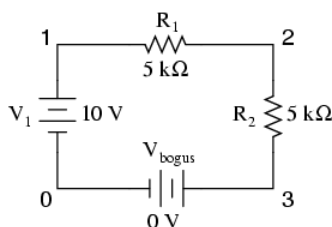
Inserting a “bogus” DC voltage source of zero volts doesn’t affect circuit operation at all, yet it provides a convenient place for SPICE to take a current measurement. Interestingly enough, it doesn’t matter that V_{bogus} is a DC source when we’re looking to measure AC current! The fact that SPICE will output an AC current reading is determined by the “ac” specification in the .print card and nothing more.

It should also be noted that the way SPICE assigns a polarity to current measurements is a bit odd. Take the following circuit as an example:



```
example v1 1 0 r1 1 2 5k r2 2 0 5k .dc v1 10 10 1 .print dc i(v1) .end
```

With 10 volts total voltage and 10 kΩ total resistance, you might expect SPICE to tell you there’s going to be 1 mA (1e-03) of current through voltage source V_1 , but in actuality, SPICE will output a figure of *negative* 1 mA (-1e-03)! SPICE regards current out of the negative end of a DC voltage source (the normal direction) to be a negative value of current rather than a positive value of current. There are times I’ll throw in a “bogus” voltage source in a DC circuit like this simply to get SPICE to output a *positive* current value:



```
example v1 1 0 r1 1 2 5k r2 2 3 5k vbogus 3 0 dc 0 .dc v1 10 10 1 .print dc i(vbogus)
```

Notice how V_{bogus} is positioned so that the circuit current will enter its positive side (node 3) and exit its negative side (node 0). This orientation will ensure a positive output figure for circuit current.

Fourier analysis

When performing a Fourier (frequency-domain) analysis on a waveform, I have found it necessary to either print or plot the waveform using the `.print` or `.plot` cards, respectively. If you don't print or plot it, SPICE will pause for a moment during analysis and then abort the job after outputting the "initial transient solution."

Also, when analyzing a square wave produced by the "pulse" source function, you must give the waveform some finite rise and fall time, or else the Fourier analysis results will be incorrect. For some reason, a perfect square wave with zero rise/fall time produces significant levels of *even* harmonics according to SPICE's Fourier analysis option, which is not true for real square waves.

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7.8: Example Circuits and Netlists

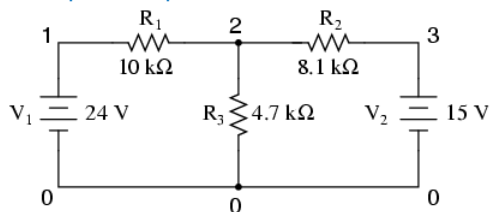
The following circuits are pre-tested netlists for SPICE 2g6, complete with short descriptions when necessary. (See Chapter 2's Computer Simulation of Electric Circuits for more information on netlists in SPICE.)

Feel free to “copy” and “paste” any of the netlists to your own SPICE source file for analysis and/or modification.

My goal here is twofold: to give practical examples of SPICE netlist design to further understanding of SPICE netlist syntax, and to show how simple and compact SPICE netlists can be in analyzing simple circuits.

All output listings for these examples have been “trimmed” of extraneous information, giving you the most succinct presentation of the SPICE output as possible. I do this primarily to save space in this document. Typical SPICE outputs contain lots of headers and summary information not necessarily germane to the task at hand. So don't be surprised when you run a simulation on your own and find that the output doesn't *exactly* look like what I have shown here!

Example multiple-source DC resistor network circuit, part 1



Without a .dc card and a .print or .plot card, the output for this netlist will only display voltages for nodes 1, 2, and 3 (with reference to node 0, of course).

Netlist:

```
Multiple dc sources v1 1 0 dc 24 v2 3 0 dc 15 r1 1 2 10k r2 2 3 8.1k r3 2 0 4.7k .end
```

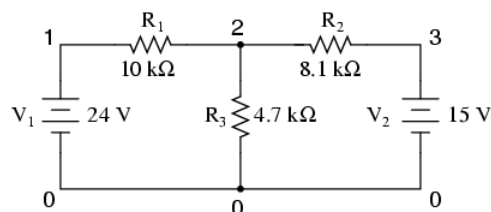
Output:

```
node voltage node voltage node voltage ( 1) 24.0000 ( 2) 9.7470 ( 3) 15.0000
```

```
voltage source currents name current v1 -1.425E-03 v2 -6.485E-04
```

```
total power dissipation 4.39E-02 watts
```

Example multiple-source DC resistor network circuit, part 2



By adding a .dc analysis card and specifying source V₁ from 24 volts to 24 volts in 1 step (in other words, 24 volts steady), we can use the .print card analysis to print out voltages between any two points we desire. Oddly enough, when the .dc analysis option is invoked, the default voltage printouts for each node (to ground) disappears, so we end up having to explicitly specify them in the .print card to see them at all.

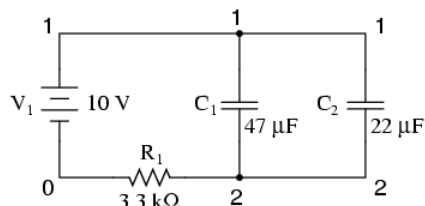
Netlist:

```
Multiple dc sources v1 1 0 v2 3 0 15 r1 1 2 10k r2 2 3 8.1k r3 2 0 4.7k .dc v1 24 24 1
```

Output:

```
v1 v(1) v(2) v(3) v(1,2) v(2,3) 2.400E+01 2.400E+01 9.747E+00 1.500E+01 1.425E+01 -5.2
```

Example RC time-constant circuit



For DC analysis, the initial conditions of any reactive component (C or L) must be specified (voltage for capacitors, current for inductors). This is provided by the last data field of each capacitor card ($ic=0$). To perform a DC analysis, the `.tran` (“transient”) analysis option must be specified, with the first data field specifying time increment in seconds, the second specifying total analysis timespan in seconds, and the “uic” telling it to “use initial conditions” when analyzing.

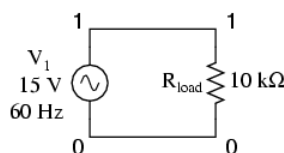
Netlist:

```
RC time delay circuit v1 1 0 dc 10 c1 1 2 47u ic=0 c2 1 2 22u ic=0 r1 2 0 3.3k .tran
```

Output:

```
time v(1,2) 0.000E+00 7.701E-06 5.000E-02 1.967E+00 1.000E-01 3.551E+00 1.500E-01 4.82
```

Plotting and analyzing a simple AC sinewave voltage circuit



This exercise does show the proper setup for plotting instantaneous values of a sine-wave voltage source with the `.plot` function (as a *transient* analysis). Not surprisingly, the Fourier analysis in this deck also requires the `.tran` (transient) analysis option to be specified over a suitable time range. The time range in this particular deck allows for a Fourier analysis with rather poor accuracy. The more cycles of the fundamental frequency that the transient analysis is performed over, the more precise the Fourier analysis will be. This is not a quirk of SPICE, but rather a basic principle of waveforms.

Netlist:

```
v1 1 0 sin(0 15 60 0 0) rload 1 0 10k * change tran card to the following for better f
```

Output:

```
time v(1) -2.000E+01 -1.000E+01 0.000E+00 1.000E+01 - - - - - - - - - - - - - - - -
```

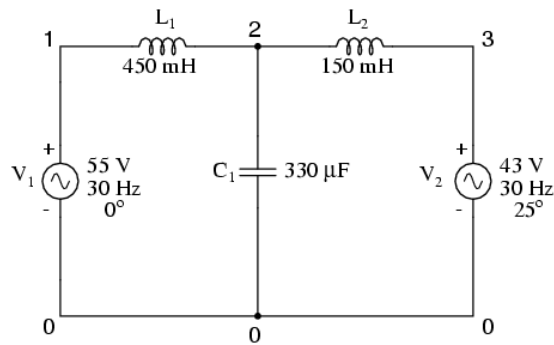
Netlist:

Output:

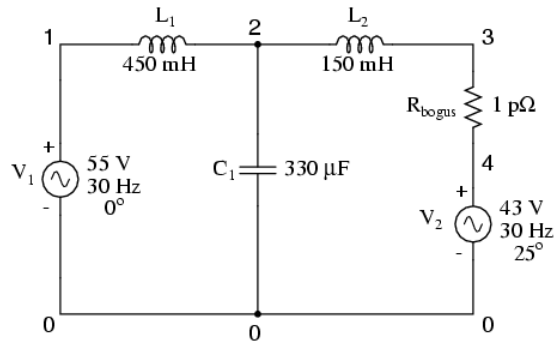
Netlist:

```
freq v(4) 1.000E-06 1.000E-04 1.000E-02 1.000E+00 - - - - - - - - - - - - - -
```

Example multiple-source AC network circuit



One of the idiosyncrasies of SPICE is its inability to handle any loop in a circuit exclusively composed of series voltage sources and inductors. Therefore, the “loop” of V_1 - L_1 - L_2 - V_2 - V_1 is unacceptable. To get around this, I had to insert a *low*-resistance resistor somewhere in that loop to break it up. Thus, we have R_{bogus} between 3 and 4 (with 1 pico-ohm of resistance), and V_2 between 4 and 0. The circuit above is the original design, while the circuit below has R_{bogus} inserted to avoid the SPICE error.



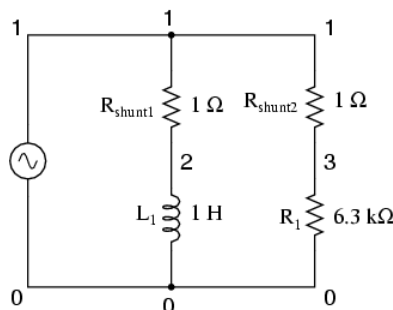
Netlist:

```
Multiple ac source v1 1 0 ac 55 0 sin v2 4 0 ac 43 25 sin l1 1 2 450m c1 2 0 330u l2 2 3 150m r_bogus 3 4 1p
```

Output:

```
freq v(2) 3.000E+01 1.413E+02
```

Example AC phase shift demonstration circuit



The currents through each leg are indicated by the voltage drops across each respective shunt resistor (1 amp = 1 volt through 1 Ω), output by the $v(1,2)$ and $v(1,3)$ terms of the .print card. The phase of the currents through each leg are indicated by the phase of the voltage drops across each respective shunt resistor, output by the $vp(1,2)$ and $vp(1,3)$ terms in the .print card.

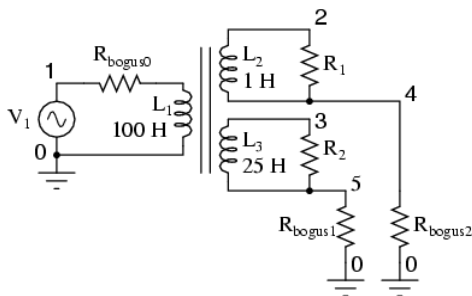
Netlist:

```
phase shift v1 1 0 ac 4 sin rshunt1 1 2 1 rshunt2 1 3 1 l1 2 0 1 r1 3 0 6.3k .ac lin 1
```

Output:

```
freq v(1,2) v(1,3) vp(1,2) vp(1,3) 1.000E+03 6.366E-04 6.349E-04 -9.000E+01 0.000E+00
```

Example transformer circuit



SPICE understands transformers as a set of mutually coupled inductors. Thus, to simulate a transformer in SPICE, you must specify the primary and secondary windings as separate inductors, then instruct SPICE to link them together with a “k” card specifying the coupling constant. For ideal transformer simulation, the coupling constant would be unity (1). However, SPICE can’t handle this value, so we use something like 0.999 as the coupling factor.

Note that *all* winding inductor pairs must be coupled with their own k cards in order for the simulation to work properly. For a two-winding transformer, a single k card will suffice. For a three-winding transformer, three k cards must be specified (to link L_1 with L_2 , L_2 with L_3 , and L_1 with L_3).

The L_1/L_2 inductance ratio of 100:1 provides a 10:1 step-down voltage transformation ratio. With 120 volts in we should see 12 volts out of the L_2 winding. The L_1/L_3 inductance ratio of 100:25 (4:1) provides a 2:1 step-down voltage transformation ratio, which should give us 60 volts out of the L_3 winding with 120 volts in.

Netlist:

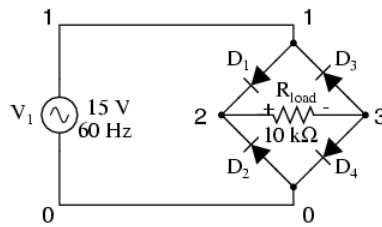
```
transformer v1 1 0 ac 120 sin rbogus0 1 6 1e-3 l1 6 0 100 l2 2 4 1 l3 3 5 25 k1 l1 l2
```

Output:

```
freq v(1) v(2) v(3) 6.000E+01 1.200E+02 1.199E+01 5.993E+01
```

In this example, R_{bogus0} is a very low-value resistor, serving to break up the source/inductor loop of V_1/L_1 . R_{bogus1} and R_{bogus2} are very high-value resistors necessary to provide DC paths to a ground on each of the isolated circuits. Note as well that one side of the primary circuit is directly grounded. Without these ground references, SPICE will produce errors!

Example full-wave bridge rectifier circuit



Diodes, like all semiconductor components in SPICE, must be modeled so that SPICE knows all the nitty-gritty details of how they're supposed to work. Fortunately, SPICE comes with a few generic models, and the diode is the most basic. Notice the `.model` card which simply specifies "d" as the generic diode model for `mod1`. Again, since we're plotting the waveforms here, we need to specify all parameters of the AC source in a single card and print/plot all values using the `.tran` option.

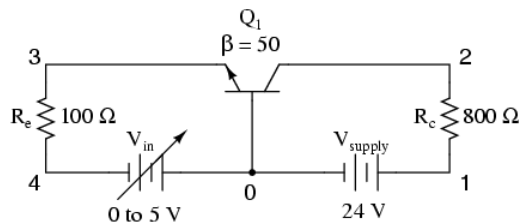
Netlist:

```
fullwave bridge rectifier v1 1 0 sin(0 15 60 0 0) rload 1 0 10k d1 1 2 mod1 d2 0 2 mod1
```

Output:

```
legend: *: v(1) +: v(2,3) time v(1) (*)----- -2.000E+01 -1.000E+01 0.000E+00 1.000E+01
```

Example common-base BJT transistor amplifier circuit



This analysis sweeps the input voltage (V_{in}) from 0 to 5 volts in 0.1 volt increments, then prints out the voltage between the collector and emitter leads of the transistor $v(2,3)$. The transistor (Q_1) is an NPN with a forward Beta of 50.

Netlist:

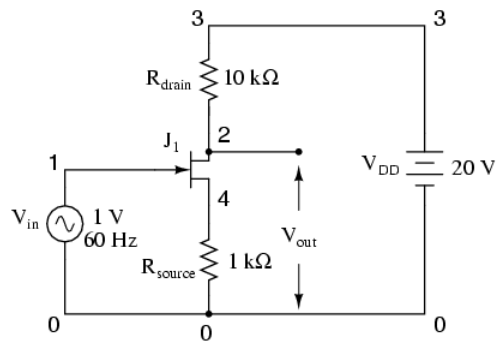
```
Common-base BJT amplifier vsupply 1 0 dc 24 vin 0 4 dc rc 1 2 800 re 3 4 100 q1 2 0 3
```

Output:

```
vin v(2,3) 0.000E+00 2.400E+01 1.000E-01 2.410E+01 2.000E-01 2.420E+01 3.000E-01 2.430E+01
```

```
vin v(2,3) 0.000E+00 1.000E+01 2.000E+01 3.000E+01 - - - - -
```


Example common-source JFET amplifier circuit with self-bias



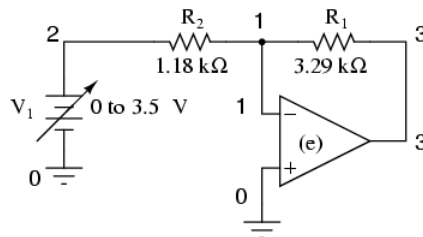
Netlist:

```
common source jfet amplifier vin 1 0 sin(0 1 60 0 0) vdd 3 0 dc 20 rdrain 3 2 10k rsou
```

Output:

```
legend: *: v(2) +: v(1) time v(2) (*)----- 1.400E+01 1.600E+01 1.800E+01 2.000E+01
```

Example inverting op-amp circuit



To simulate an ideal operational amplifier in SPICE, we use a voltage-dependent voltage source as a differential amplifier with extremely high gain. The “e” card sets up the dependent voltage source with four nodes, 3 and 0 for voltage output, and 1 and 0 for voltage input. No power supply is needed for the dependent voltage source, unlike a real operational amplifier. The voltage gain is set at 999,000 in this case. The input voltage source (V_1) sweeps from 0 to 3.5 volts in 0.05 volt steps.

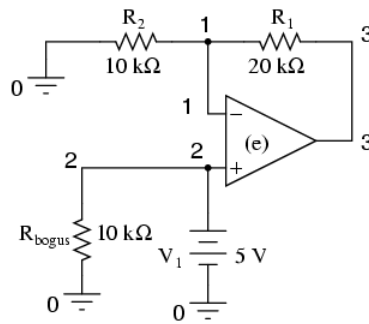
Netlist:

```
Inverting opamp v1 2 0 dc e 3 0 0 1 999k r1 3 1 3.29k r2 1 2 1.18k .dc v1 0 3.5 0.05
```

Output:

```
v1 v(3) 0.000E+00 0.000E+00 5.000E-02 -1.394E-01 1.000E-01 -2.788E-01 1.500E-01 -4.182
```

Example noninverting op-amp circuit



Another example of a SPICE quirk: since the dependent voltage source “e” isn’t considered a load to voltage source V_1 , SPICE interprets V_1 to be open-circuited and will refuse to analyze it. The fix is to connect R_{bogus} in parallel with V_1 to act as a DC load. Being directly connected across V_1 , the resistance of R_{bogus} is not crucial to the operation of the circuit, so 10 kΩ will work fine. I decided not to sweep the V_1 input voltage at all in this circuit for the sake of keeping the netlist and output listing simple.

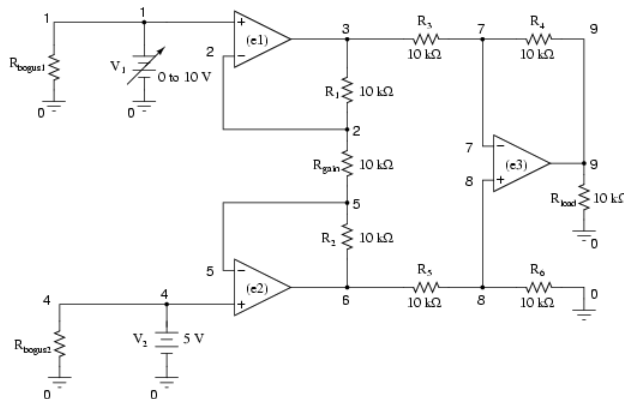
Netlist:

```
noninverting opamp v1 2 0 dc 5 rbogus 2 0 10k e 3 0 2 1 999k r1 3 1 20k r2 1 0 10k .er
```

Output:

```
node voltage node voltage node voltage ( 1) 5.0000 ( 2) 5.0000 ( 3) 15.0000
```

Example instrumentation amplifier circuit



Note the very high-resistance R_{bogus1} and R_{bogus2} resistors in the netlist (not shown in schematic for brevity) across each input voltage source, to keep SPICE from thinking V_1 and V_2 were open-circuited, just like the other op-amp circuit examples.

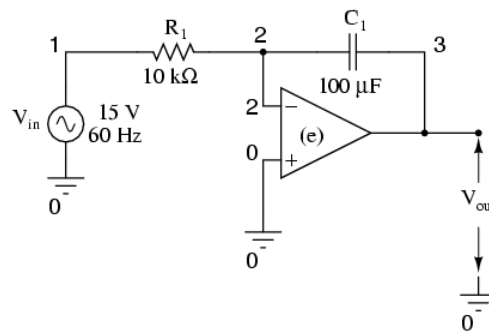
Netlist:

```
Instrumentation amplifier v1 1 0 rbogus1 1 0 9e12 v2 4 0 dc 5 rbogus2 4 0 9e12 e1 3 0
```

Output:

```
v1 v(9) v(3,6) 0.000E+00 1.500E+01 -1.500E+01 1.000E+00 1.200E+01 -1.200E+01 2.000E+00
```

Example op-amp integrator circuit with sinewave input



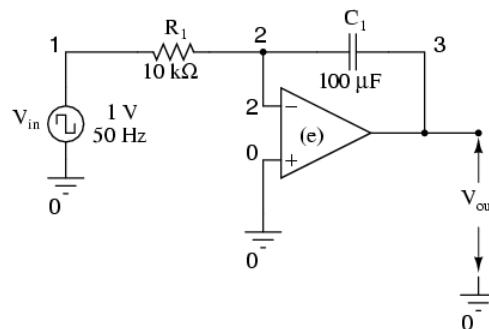
Netlist:

```
Integrator with sinewave input vin 1 0 sin (0 15 60 0 0) r1 1 2 10k c1 2 3 150u ic=0 e
```

Output:

```
legend: *: v(1) +: v(3) time v(1) (*)----- -2.000E+01 -1.000E+01 0.000E+00 1.000E+00
```

Example op-amp integrator circuit with squarewave input



Netlist:

```
Integrator with squarewave input vin 1 0 pulse (-1 1 0 0 0 10m 20m) r1 1 2 1k c1 2 3 1
```

Output:

```
<(1) +: v(3) time v(1) (*)-----1.000E+00 -5.000E-01 0.000E+00 5.000E-01 1.000E+00 (+)-----1.000E-01 -5.000E-02
0.000E+00 5.000E-02 1.000E-01 ----- 0.000E+00 -1.000E+00 * . + . 1.000E-03
1.000E+00 . . + . * 2.000E-03 1.000E+00 . . + . * 3.000E-03 1.000E+00 . . + . * 4.000E-03 1.000E+00 . . + . * 5.000E-03
1.000E+00 . . + . * 6.000E-03 1.000E+00 . . + . * 7.000E-03 1.000E+00 . . + . * 8.000E-03 1.000E+00 . . + . * 9.000E-03
1.000E+00 . . + . * 1.000E-02 1.000E+00 . . + . * 1.100E-02 1.000E+00 . . + . * 1.200E-02 -1.000E+00 * + . . . 1.300E-02
-1.000E+00 * + . . . 1.400E-02 -1.000E+00 * + . . . 1.500E-02 -1.000E+00 * + . . . 1.600E-02 -1.000E+00 * + . . . 1.700E-02
-1.000E+00 * + . . . 1.800E-02 -1.000E+00 * + . . . 1.900E-02 -1.000E+00 * + . . . 2.000E-02 -1.000E+00 * + . . . 2.100E-02
1.000E+00 . . + . * 2.200E-02 1.000E+00 . . + . * 2.300E-02 1.000E+00 . . + . * 2.400E-02 1.000E+00 . . + . * 2.500E-02
1.000E+00 . . + . * 2.600E-02 1.000E+00 . . + . * 2.700E-02 1.000E+00 . . + . * 2.800E-02 1.000E+00 . . + . * 2.900E-02
1.000E+00 . . + . * 3.000E-02 1.000E+00 . . + . * 3.100E-02 1.000E+00 . . + . * 3.200E-02 -1.000E+00 * + . . . 3.300E-02
-1.000E+00 * + . . . 3.400E-02 -1.000E+00 * + . . . 3.500E-02 -1.000E+00 * + . . . 3.600E-02 -1.000E+00 * + . . . 3.700E-02
-1.000E+00 * + . . . 3.800E-02 -1.000E+00 * + . . . 3.900E-02 -1.000E+00 * + . . . 4.000E-02 -1.000E+00 * + . . . 4.100E-02
```

1.000E+00 . . + . . * 4.200E-02 1.000E+00 . . + . . * 4.300E-02 1.000E+00 . . + . . * 4.400E-02 1.000E+00 . . + . . * 4.500E-02
1.000E+00 . . + . . * 4.600E-02 1.000E+00 . . + . . * 4.700E-02 1.000E+00 . . + . . * 4.800E-02 1.000E+00 . . + . . * 4.900E-02
1.000E+00 . . + . . * 5.000E-02 1.000E+00 + . . * -----

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CHAPTER OVERVIEW

8: Troubleshooting -- Theory And Practice

- 8.1: Questions to Ask Before Proceeding
- 8.2: General Troubleshooting Tips
- 8.3: Specific Troubleshooting Techniques
- 8.4: Likely Failures in Proven Systems
- 8.5: Likely Failures in Unproven Systems
- 8.6: Potential Pitfalls

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8.1: Questions to Ask Before Proceeding

- Has the system ever worked before? If yes, has anything happened to it since then that could cause the problem?
- Has this system proven itself to be prone to certain types of failure?
- How urgent is the need for repair?
- What are the *safety concerns*, before I start troubleshooting?
- What are the process quality concerns, before I start troubleshooting (what can I do without causing interruptions in production)?

These preliminary questions are not trivial. Indeed, they are essential to expedient and safe troubleshooting. They are especially important when the system to be trouble-shot is large, dangerous, and/or expensive.

Sometimes the troubleshooter will be required to work on a system that is still in full operation (perhaps the ultimate example of this is a doctor diagnosing a live patient). Once the cause or causes are determined to a high degree of certainty, there is the step of corrective action. Correcting a system fault without significantly interrupting the operation of the system can be very challenging, and it deserves thorough planning.

When there is high risk involved in taking corrective action, such as is the case with performing surgery on a patient or making repairs to an operating process in a chemical plant, it is essential for the worker(s) to plan ahead for possible trouble. One question to ask before proceeding with repairs is, “how and at what point(s) can I abort the repairs if something goes wrong?” In risky situations, it is vital to have planned “escape routes” in your corrective action, just in case things do not go as planned. A surgeon operating on a patient knows if there are any “points of no return” in such a procedure, and stops to re-check the patient before proceeding past those points. He or she also knows how to “back out” of a surgical procedure at those points if needed.

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8.2: General Troubleshooting Tips

When first approaching a failed or otherwise misbehaving system, the new troubleshooter often doesn't know where to begin. The following strategies are not exhaustive by any means, but provide the troubleshooter with a simple checklist of questions to ask in order to start isolating the problem.

As for tips, these troubleshooting suggestions are not comprehensive procedures: they serve as starting points only for the troubleshooting process. An essential part of expedient troubleshooting is probability assessment, and these tips help the troubleshooter determine which possible points of failure are more or less likely than others. Final isolation of the system failure is usually determined through more specific techniques (outlined in the next section—*Specific Troubleshooting Techniques*).

Prior occurrence

If this device or process has been historically known to fail in a certain particular way, and the conditions leading to this common failure have not changed, check for this “way” first. A corollary to this troubleshooting tip is the directive to keep detailed records of failure. Ideally, a computer-based failure log is optimal, so that failures may be referenced by and correlated to a number of factors such as time, date, and environmental conditions.

Example: *The car's engine is overheating. The last two times this happened, the cause was low coolant level in the radiator.*

What to do: Check the coolant level first. Of course, past history by no means guarantees the present symptoms are caused by the same problem, but since this is more likely, it makes sense to check this first.

If, however, the cause of routine failure in a system has been corrected (i.e. the leak causing low coolant level in the past has been repaired), then this may not be a probable cause of trouble this time.

Recent alterations

If a system has been having problems immediately after some kind of maintenance or other change, the problems might be linked to those changes.

Example: *The mechanic recently tuned my car's engine, and now I hear a rattling noise that I didn't hear before I took the car in for repair.*

What to do: Check for something that may have been left loose by the mechanic after his or her tune-up work.

Function vs. non-function

If a system isn't producing the desired end result, look for what it is doing correctly; in other words, identify where the problem is *not*, and focus your efforts elsewhere. Whatever components or subsystems necessary for the properly working parts to function are probably okay. The degree of fault can often tell you what part of it is to blame.

Example: *The radio works fine on the AM band, but not on the FM band.*

What to do: Eliminate from the list of possible causes, anything in the radio necessary for the AM band's function. Whatever the source of the problem is, it is specific to the FM band and not to the AM band. This eliminates the audio amplifier, speakers, fuse, power supply, and almost all external wiring. Being able to eliminate sections of the system as possible failures reduces the scope of the problem and makes the rest of the troubleshooting procedure more efficient.

Hypothesize

Based on your knowledge of how a system works, think of various kinds of failures that would cause this problem (or these phenomena) to occur, and check for those failures (starting with the most likely based on circumstances, history, or knowledge of component weaknesses).

Example: *The car's engine is overheating.*

What to do: Consider possible causes for overheating, based on what you know of engine operation. Either the engine is generating too much heat, or not getting rid of the heat well enough (most likely the latter). Brainstorm some possible causes: a loose fan belt, clogged radiator, bad water pump, low coolant level, etc. Investigate each one of those possibilities before investigating alternatives.

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8.3: Specific Troubleshooting Techniques

After applying some of the general troubleshooting tips to narrow the scope of a problem's location, there are techniques useful in further isolating it. Here are a few:

Swap identical components

In a system with identical or parallel subsystems, swap components between those subsystems and see whether or not the problem moves with the swapped component. If it does, you've just swapped the faulty component; if it doesn't, keep searching!

This is a powerful troubleshooting method, because it gives you both a positive and a negative indication of the swapped component's fault: when the bad part is exchanged between identical systems, the formerly broken subsystem will start working again and the formerly good subsystem will fail.

I was once able to troubleshoot an elusive problem with an automotive engine ignition system using this method: I happened to have a friend with an automobile sharing the exact same model of ignition system. We swapped parts between the engines (distributor, spark plug wires, ignition coil—one at a time) until the problem moved to the other vehicle. The problem happened to be a “weak” ignition coil, and it only manifested itself under heavy load (a condition that could not be simulated in my garage). Normally, this type of problem could only be pinpointed using an ignition system analyzer (or oscilloscope) *and* a dynamometer to simulate loaded driving conditions. This technique, however, confirmed the source of the problem with 100% accuracy, using no diagnostic equipment whatsoever.

Occasionally you may swap a component and find that the problem still exists, but has changed in some way. This tells you that the components you just swapped are *somehow different* (different calibration, different function), and nothing more. However, don't dismiss this information just because it doesn't lead you straight to the problem—look for other changes in the system as a whole as a result of the swap, and try to figure out what these changes tell you about the source of the problem.

An important caveat to this technique is the possibility of causing further damage. Suppose a component has failed because of another, less conspicuous failure in the system. Swapping the failed component with a good component will cause the good component to fail as well. For example, suppose that a circuit develops a short, which “blows” the protective fuse for that circuit. The blown fuse is not evident by inspection, and you don't have a meter to electrically test the fuse, so you decide to swap the suspect fuse with one of the same rating from a working circuit. As a result of this, the good fuse that you move to the shorted circuit blows as well, leaving you with two blown fuses and two non-working circuits. At least you know for certain that the original fuse was blown, because the circuit it was moved to stopped working after the swap, but this knowledge was gained only through the loss of a good fuse and the additional “down time” of the second circuit.

Another example to illustrate this caveat is the ignition system problem previously mentioned. Suppose that the “weak” ignition coil had caused the engine to backfire, damaging the muffler. If swapping ignition system components with another vehicle causes the problem to move to the other vehicle, damage may be done to the other vehicle's muffler as well. As a general rule, the technique of swapping identical components should be used only when there is minimal chance of causing additional damage. It is an excellent technique for isolating non-destructive problems.

Example 1: *You're working on a CNC machine tool with X, Y, and Z-axis drives. The Y axis is not working, but the X and Z axes are working. All three axes share identical components (feedback encoders, servo motor drives, servo motors).*

What to do: Exchange these identical components, one at a time, Y axis and either one of the working axes (X or Z), and see after each swap whether or not the problem has moved with the swap.

Example 2: *A stereo system produces no sound on the left speaker, but the right speaker works just fine.*

What to do: Try swapping respective components between the two channels and see if the problem changes sides, from left to right. When it does, you've found the defective component. For instance, you could swap the speakers between channels: if the problem moves to the other side (i.e. the same speaker that was dead before is still dead, now that its connected to the right channel cable) then you know that speaker is bad. If the problem stays on the same side (i.e. the speaker formerly silent is now producing sound after having been moved to the other side of the room and connected to the other cable), then you know the speakers are fine, and the problem must lie somewhere else (perhaps in the cable connecting the silent speaker to the amplifier, or in the amplifier itself).

If the speakers have been verified as good, then you could check the cables using the same method. Swap the cables so that each one now connects to the other channel of the amplifier and to the other speaker. Again, if the problem changes sides (i.e. now the right speaker is now “dead” and the left speaker now produces sound), then the cable now connected to the right speaker must be defective. If neither swap (the speakers nor the cables) causes the problem to change sides from left to right, then the problem must lie within the amplifier (i.e. the left channel output must be “dead”).

Remove parallel components

If a system is composed of several parallel or redundant components which can be removed without crippling the whole system, start removing these components (one at a time) and see if things start to work again.

Example 1: A “star” topology communications network between several computers has failed. None of the computers are able to communicate with each other.

What to do: Try unplugging the computers, one at a time from the network, and see if the network starts working again after one of them is unplugged. If it does, then that last unplugged computer may be the one at fault (it may have been “jamming” the network by constantly outputting data or noise).

Example 2: A household fuse keeps blowing (or the breaker keeps tripping open) after a short amount of time.

What to do: Unplug appliances from that circuit until the fuse or breaker quits interrupting the circuit. If you can eliminate the problem by unplugging a single appliance, then that appliance might be defective. If you find that unplugging almost any appliance solves the problem, then the circuit may simply be overloaded by too many appliances, neither of them defective.

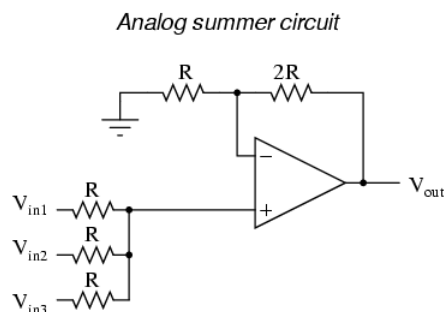
Divide system into sections and test those sections

In a system with multiple sections or stages, carefully measure the variables going in and out of each stage until you find a stage where things don’t look right.

Example 1: A radio is not working (producing no sound at the speaker))

What to do: Divide the circuitry into stages: tuning stage, mixing stages, amplifier stage, all the way through to the speaker(s). Measure signals at test points between these stages and tell whether or not a stage is working properly.

Example 2: An analog summer circuit is not functioning properly.



What to do: I would test the passive averager network (the three resistors at the lower-left corner of the schematic) to see that the proper (averaged) voltage was seen at the noninverting input of the op-amp. I would then measure the voltage at the inverting input to see if it was the same as at the noninverting input (or, alternatively, measure the voltage difference between the two inputs of the op-amp, as it should be zero). Continue testing sections of the circuit (or just test points within the circuit) to see if you measure the expected voltages and currents.

Simplify and rebuild

Closely related to the strategy of dividing a system into sections, this is actually a design and fabrication technique useful for new circuits, machines, or systems. It’s always easier begin the design and construction process in little steps, leading to larger and larger steps, rather than to build the whole thing at once and try to troubleshoot it as a whole.

Suppose that someone were building a custom automobile. He or she would be foolish to bolt all the parts together without checking and testing components and subsystems as they went along, expecting everything to work perfectly after its all assembled.

Ideally, the builder would check the proper operation of components along the way through the construction process: start and tune the engine *before* its connected to the drivetrain, check for wiring problems *before* all the cover panels are put in place, check the brake system in the driveway *before* taking it out on the road, etc.

Countless times I've witnessed students build a complex experimental circuit and have trouble getting it to work because they didn't stop to check things along the way: test all resistors *before* plugging them into place, make sure the power supply is regulating voltage adequately *before* trying to power anything with it, etc. It is human nature to rush to completion of a project, thinking that such checks are a waste of valuable time. However, more time will be wasted in troubleshooting a malfunctioning circuit than would be spent checking the operation of subsystems throughout the process of construction.

Take the example of the analog summer circuit in the previous section for example: what if it wasn't working properly? How would you simplify it and test it in stages? Well, you could reconnect the op-amp as a basic comparator and see if its responsive to differential input voltages, and/or connect it as a voltage follower (buffer) and see if it outputs the same analog voltage as what is input. If it doesn't perform these simple functions, it will never perform its function in the summer circuit! By stripping away the complexity of the summer circuit, paring it down to an (almost) bare op-amp, you can test that component's functionality and then build from there (add resistor feedback and check for voltage amplification, then add input resistors and check for voltage summing), checking for expected results along the way.

Trap a signal

Set up instrumentation (such as a datalogger, chart recorder, or multimeter set on "record" mode) to monitor a signal over a period of time. This is especially helpful when tracking down intermittent problems, which have a way of showing up the moment you've turned your back and walked away.

This may be essential for proving what happens first in a fast-acting system. Many fast systems (especially shutdown "trip" systems) have a "first out" monitoring capability to provide this kind of data.

Example #1: *A turbine control system shuts automatically in response to an abnormal condition. By the time a technician arrives at the scene to survey the turbine's condition, however, everything is in a "down" state and its impossible to tell what signal or condition was responsible for the initial shutdown, as all operating parameters are now "abnormal."*

What to do: One technician I knew used a video camera to record the turbine control panel, so he could see what happened (by indications on the gauges) first in an automatic-shutdown event. Simply by looking at the panel after the fact, there was no way to tell *which* signal shut the turbine down, but the videotape playback would show what happened in sequence, down to a frame-by-frame time resolution.

Example #2: *An alarm system is falsely triggering, and you suspect it may be due to a specific wire connection going bad. Unfortunately, the problem never manifests itself while you're watching it!*

What to do: Many modern digital multimeters are equipped with "record" settings, whereby they can monitor a voltage, current, or resistance over time and note whether that measurement deviates substantially from a regular value. This is an invaluable tool for use in "intermittent" electronic system failures.

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8.4: Likely Failures in Proven Systems

The following problems are arranged in order from most likely to least likely, top to bottom. This order has been determined largely from personal experience troubleshooting electrical and electronic problems in automotive, industrial, and home applications. This order also assumes a circuit or system that has been proven to function as designed and has failed after substantial operation time. Problems experienced in newly assembled circuits and systems do not necessarily exhibit the same probabilities of occurrence.

Operator error

A frequent cause of system failure is error on the part of those human beings operating it. This cause of trouble is placed at the top of the list, but of course, the actual likelihood depends largely on the particular individuals responsible for operation. When operator error is the cause of a failure, it is *unlikely* that it will be admitted prior to investigation. I do not mean to suggest that operators are incompetent and irresponsible—quite the contrary: these people are often your best teachers for learning system function and obtaining a history of failure—but the reality of human error cannot be overlooked. A positive attitude coupled with good interpersonal skills on the part of the troubleshooter goes a long way in troubleshooting when human error is the root cause of failure.

Bad wire connections

As incredible as this may sound to the new student of electronics, a high percentage of electrical and electronic system problems are caused by a very simple source of trouble: poor (i.e. open or shorted) wire connections. This is especially true when the environment is hostile, including such factors as high vibration and/or a corrosive atmosphere. Connection points found in any variety of plug-and-socket connector, terminal strip, or splice are at the greatest risk for failure. The category of “connections” also includes mechanical switch contacts, which can be thought of as a high-cycle connector. Improper wire termination lugs (such as a compression-style connector crimped on the end of a

solid wire—a definite *faux pas*) can cause high-resistance connections after a period of trouble-free service.

It should be noted that connections in low-voltage systems tend to be far more troublesome than connections in high-voltage systems. The main reason for this is the effect of arcing across a discontinuity (circuit break) in higher-voltage systems tends to blast away insulating layers of dirt and corrosion, and may even weld the two ends together if sustained long enough. Low-voltage systems tend not to generate such vigorous arcing across the gap of a circuit break, and also tend to be more sensitive to additional resistance in the circuit. Mechanical switch contacts used in low-voltage systems benefit from having the recommended minimum *wetting current* conducted through them to promote a healthy amount of arcing upon opening, even if this level of current is not necessary for the operation of other circuit components.

Although *open* failures tend to be more common than *shorted* failures, “shorts” still constitute a substantial percentage of wiring failure modes. Many shorts are caused by degradation of wire insulation. This, again, is especially true when the environment is hostile, including such factors as high vibration, high heat, high humidity, or high voltage. It is rare to find a mechanical switch contact that is failed shorted, except in the case of high-current contacts where contact “welding” may occur in over current conditions. Shorts may also be caused by conductive build up across terminal strip sections or the backs of printed circuit boards.

A common case of shorted wiring is the *ground fault*, where a conductor accidentally makes contact with either earth or chassis ground. This may change the voltage(s) present between other conductors in the circuit and ground, thereby causing bizarre system malfunctions and/or personnel hazard.

Power supply problems

These generally consist of tripped overcurrent protection devices or damage due to overheating. Although power supply circuitry is usually less complex than the circuitry being powered and therefore should figure to be less prone to failure on that basis alone, it generally handles more power than any other portion of the system and therefore must deal with greater voltages and/or currents. Also, because of its relative design simplicity, a system’s power supply may not receive the engineering attention it deserves, most of the engineering focus devoted to more glamorous parts of the system.

Active components

Active components (amplification devices) tend to fail with greater regularity than passive (non-amplifying) devices, due to their greater complexity and tendency to amplify overvoltage/overcurrent conditions. Semiconductor devices are notoriously prone to failure due to electrical transient (voltage/current surge) overloading and thermal (heat) overloading. Electron tube devices are far more resistant to both of these failure modes but are generally more prone to mechanical failures due to their fragile construction.

Passive components

Non-amplifying components are the most rugged of all, their relative simplicity granting them a statistical advantage over active devices. The following list gives an approximate relation of failure probabilities (again, top being the most likely and bottom being the least likely):

- Capacitors (shorted), especially *electrolytic* capacitors. The paste electrolyte tends to lose moisture with age, leading to failure. Thin dielectric layers may be punctured by overvoltage transients.
- Diodes open (rectifying diodes) or shorted (Zener diodes).
- Inductor and transformer windings open or shorted to conductive core. Failures related to overheating (insulation breakdown) are easily detected by smell.
- Resistors open, almost never shorted. Usually, this is due to overcurrent heating, although it is less frequently caused by overvoltage transient (arc-over) or physical damage (vibration or impact). Resistors may also change resistance value if overheated!

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8.5: Likely Failures in Unproven Systems

“All men are liable to error;”

—*John Locke*

Whereas the last section deals with component failures in systems that have been successfully operating for some time, this section concentrates on the problems plaguing brand-new systems. In this case, failure modes are generally not of the aging kind but are related to mistakes in design and assembly caused by human beings.

Wiring problems

In this case, bad connections are usually due to assembly error, such as connection to the wrong point or poor connector fabrication. Shorted failures are also seen, but usually, involve misconnections (conductors inadvertently attached to grounding points) or wires pinched under box covers.

Another wiring-related problem seen in new systems is that of electrostatic or electromagnetic interference between different circuits by way of close wiring proximity. This kind of problem is easily created by routing sets of wires too close to each other (especially routing signal cables close to power conductors) and tends to be very difficult to identify and locate with test equipment.

Power supply problems

Blown fuses and tripped circuit breakers are likely sources of trouble, especially if the project in question is an addition to an already-functioning system. Loads may be larger than expected, resulting in overloading and subsequent failure of power supplies.

Defective components

In the case of a newly-assembled system, component fault probabilities are not as predictable as in the case of an operating system that fails with age. Any type of component—active or passive—may be found defective or of imprecise value “out of the box” with roughly equal probability, barring any specific sensitivities in shipping (i.e. fragile vacuum tubes or electrostatically sensitive semiconductor components). Moreover, these types of failures are not always as easy to identify by sight or smell as an age- or transient-induced failure.

Improper system configuration

Increasingly seen in large systems using microprocessor-based components, “programming” issues can still plague non-microprocessor systems in the form of incorrect time-delay relay settings, limit switch calibrations, and drum switch sequences. Complex components having configuration “jumpers” or switches to control behavior may not be “programmed” properly.

Components may be used in a new system outside of their tolerable ranges. Resistors, for example, with too low of power ratings, or too great of tolerance, may have been installed. Sensors, instruments, and controlling mechanisms may be uncalibrated, or calibrated to the wrong ranges.

Design error

Perhaps the most difficult to pinpoint and the slowest to be recognized (especially by the chief designer) is the problem of design error, where the system fails to function simply because it *cannot* function as designed. This may be as trivial as the designer specifying the wrong components in a system, or as fundamental as a system not working due to the designer’s improper knowledge of physics.

I once saw a turbine control system installed that used a low-pressure switch on the lubrication oil tubing to shut down the turbine if oil pressure dropped to an insufficient level. The oil pressure for lubrication was supplied by an oil pump turned by the turbine. When installed, the turbine refused to start. Why? Because when it was stopped, the oil pump was not turning, thus there was no oil pressure to lubricate the turbine. The low-oil-pressure switch detected this condition and the control system maintained the turbine in shutdown mode, preventing it from starting. This is a classic example of a design flaw, and it could only be corrected by a change in the system logic.

While most design flaws manifest themselves early in the operational life of the system, some remain hidden until just the right conditions exist to trigger the fault. These types of flaws are the most difficult to uncover, as the troubleshooter usually overlooks the possibility of design error due to the fact that the system is assumed to be “proven.” The example of the turbine lubrication system was a design flaw impossible to ignore on start-up. An example of a “hidden” design flaw might be a faulty emergency coolant system for a machine, designed to remain inactive until certain abnormal conditions are reached—conditions which might never be experienced in the life of the system.

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8.6: Potential Pitfalls

Fallacious reasoning and poor interpersonal relations account for more failed or belabored troubleshooting efforts than any other impediments. With this in mind, the aspiring troubleshooter needs to be familiar with a few common troubleshooting mistakes.

Trusting that a brand-new component will always be good. While it is generally true that a new component will be in good condition, it is not *always* true. It is also possible that a component has been mislabeled and may have the wrong value (usually this mislabeling is a mistake made at the point of distribution or warehousing and not at the manufacturer, but again, *not always*!).

Not periodically checking your test equipment. This is especially true with battery-powered meters, as weak batteries may give spurious readings. When using meters to safety-check for dangerous voltage, remember to test the meter on a known source of voltage both *before* and *after* checking the circuit to be serviced, to make sure the meter is in proper operating condition.

Assuming there is only one failure to account for the problem. Single-failure system problems are ideal for troubleshooting, but sometimes failures come in multiple numbers. In some instances, the failure of one component may lead to a system condition that damages other components. Sometimes a component in marginal condition goes undetected for a long time, then when another component fails the system suffers from problems with *both* components.

Mistaking coincidence for causality. Just because two events occurred at nearly the same time does *not* necessarily mean one event *caused* the other! They may be both consequences of a common cause, or they may be totally unrelated! If possible, try to duplicate the same condition suspected to be the cause and see if the event suspected to be the coincidence happens again. If not, then there is either no causal relationship as assumed. This may mean there is no causal relationship between the two events whatsoever, or that there is a causal relationship, but just not the one you expected.

Self-induced blindness. After a long effort at troubleshooting a difficult problem, you may become tired and begin to overlook crucial clues to the problem. Take a break and let someone else look at it for a while. You will be amazed at what a difference this can make. On the other hand, it is generally a bad idea to solicit help at the start of the troubleshooting process. Effective troubleshooting involves complex, multi-level thinking, which is not easily communicated with others. More often than not, “team troubleshooting” takes more time and causes more frustration than doing it yourself. An exception to this rule is when the knowledge of the troubleshooters is complementary: for example, a technician who knows electronics but not machine operation teamed with an operator who knows machine function but not electronics.

Failing to question the troubleshooting work of others on the same job. This may sound rather cynical and misanthropic, but it is sound scientific practice. Because it is easy to overlook important details, troubleshooting data received from another troubleshooter should be personally verified before proceeding. This is a common situation when troubleshooters “change shifts” and a technician takes over for another technician who is leaving before the job is done. It is important to exchange information, but do not assume the prior technician checked everything they said they did or checked it perfectly. I’ve been hindered in my troubleshooting efforts on many occasions by failing to verify what someone else told me they checked.

Being pressured to “hurry up.” When an important system fails, there will be pressure from other people to fix the problem as quickly as possible. As they say in business, “time is money.” Having been on the receiving end of this pressure many times, I can understand the need for expedience. However, in many cases, there is a higher priority: caution. If the system in question harbors great danger to life and limb, the pressure to “hurry up” may result in injury or death. At the very least, hasty repairs may result in further damage when the system is restarted. Most failures can be recovered or at least temporarily repaired in short time if approached intelligently. Improper “fixes” resulting in haste often lead to damage that *cannot* be recovered in short time, if ever. If the potential for greater harm is present, the troubleshooter needs to politely address the pressure received from others, and maintain their perspective in the midst of chaos. Interpersonal skills are just as important in this realm as technical ability!

Finger-pointing. It is all too easy to blame a problem on someone else, for reasons of ignorance, pride, laziness, or some other unfortunate facet of human nature. When the responsibility for system maintenance is divided into departments or work crews, troubleshooting efforts are often hindered by blame cast between groups. “It’s a mechanical problem . . . it’s an electrical problem . . . it’s an instrument problem . . .” ad infinitum, ad nauseum, is all too common in the workplace. I have found that a positive attitude does more to quench the fires of the blame than anything else.

On one particular job, I was summoned to fix a problem in a hydraulic system assumed to be related to the electronic metering and controls. My troubleshooting isolated the source of trouble to a faulty control valve, which was the domain of the millwright (mechanical) crew. I knew that the millwright on shift was a contentious person, so I expected trouble if I simply passed the problem on to his department. Instead, I politely explained to him and his supervisor the nature of the problem as well as a brief synopsis of my reasoning, then proceeded to help him replace the faulty valve, even though it wasn't "my" responsibility to do so. As a result, the problem was fixed very quickly, and I gained the respect of the millwright.

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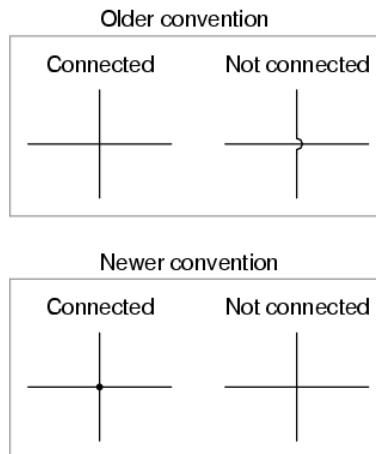
CHAPTER OVERVIEW

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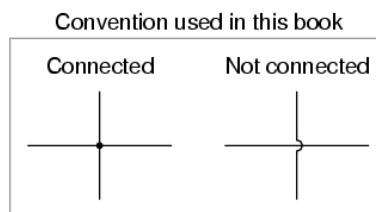
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9.1: Wires and Connections



Older electrical schematics showed connecting wires crossing, while non-connecting wires “jumped” over each other with little half-circle marks. Newer electrical schematics show connecting wires joining with a dot, while non-connecting wires cross with no dot. However, some people still use the older convention of connecting wires crossing with no dot, which may create confusion.

For this reason, I opt to use a hybrid convention, with connecting wires unambiguously connected by a dot, and non-connecting wires unambiguously “jumping” over one another with a half-circle mark. While this may be frowned upon by some, it leaves no room for interpretational error: in each case, the intent is clear and unmistakable:



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9.2: Power Sources

DC voltage



DC voltage



AC voltage



Variable
DC voltage



*A diagonal arrow
represents variability
for **any** component!*

DC current



Generator

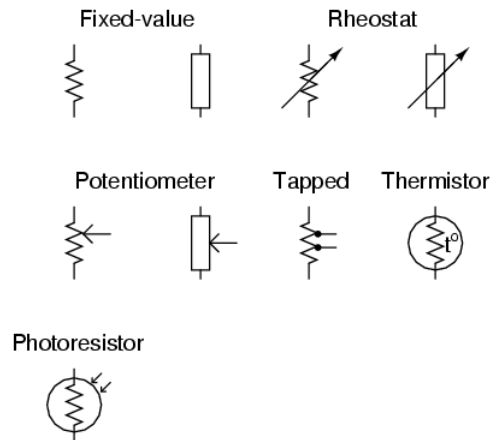


AC current



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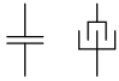
9.3: Resistors Types



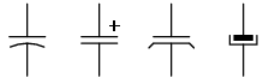
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9.4: Capacitor Types

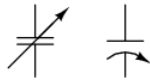
Non-polarized



Polarized (top positive)



Variable



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9.5: Inductors

Fixed-value



Iron core



Variable



Variac

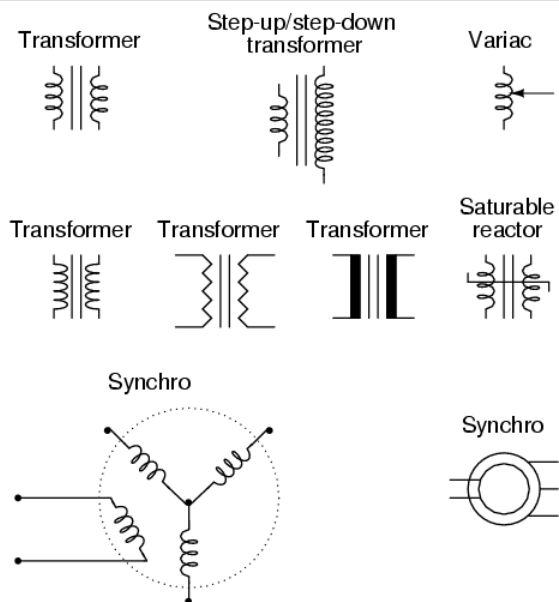


Tapped



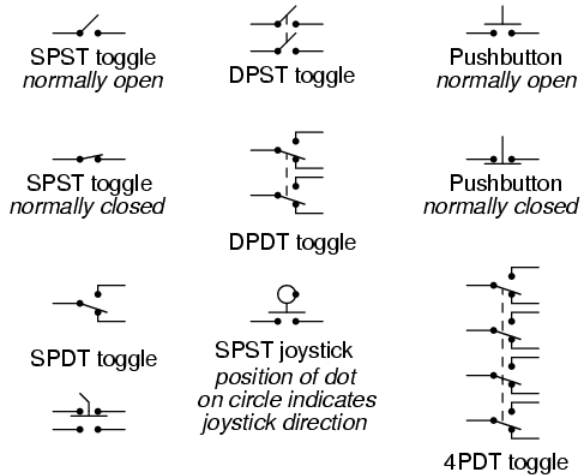
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9.6: Mutual Inductors



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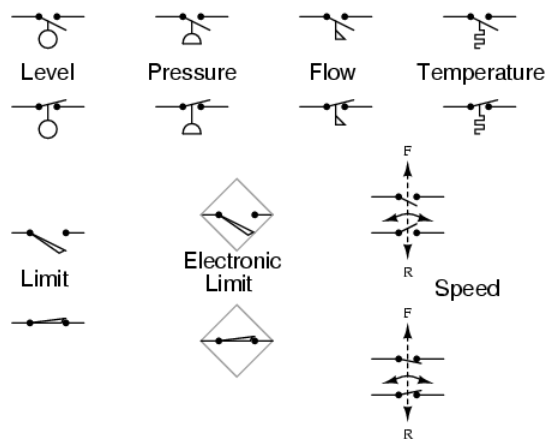
9.7: Switches, Hand Actuated



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9.8: Switches, Process Actuated

Normally open shown on top; normally closed on bottom



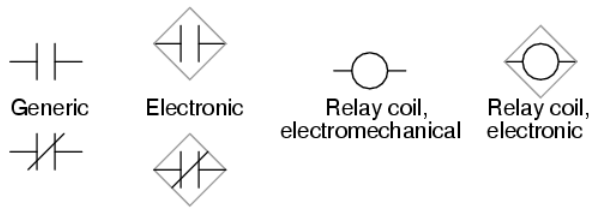
It is very important to keep in mind that the “normal” contact status of a process-actuated switch refers to its status when the process is absent and/or inactive, *not* “normal” in the sense of process conditions as expected during routine operation. For instance, a *normally-closed* low-flow detection switch installed on a coolant pipe will be maintained in the actuated state (open) when there is regular coolant flow through the pipe. If the coolant flow stops, the flow switch will go to its “normal” (un-actuated) status of closed.

A *limit* switch is one actuated by contact with a moving machine part. An *electronic limit* switch senses mechanical motion but does so using light, magnetic fields, or other non-contact means.

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9.9: Switches, Electrically Actuated (Relays)

Relay components, "ladder logic" notation style

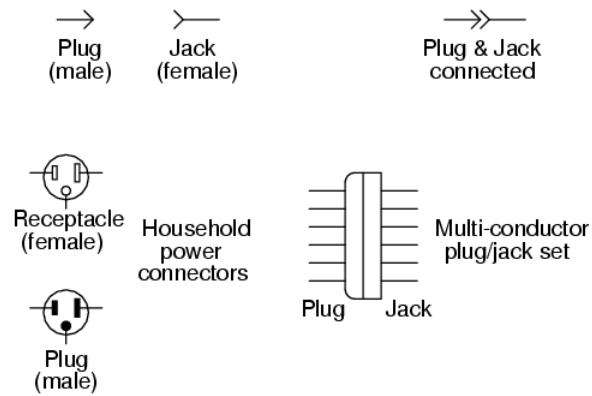


Relays, electronic schematic notation style



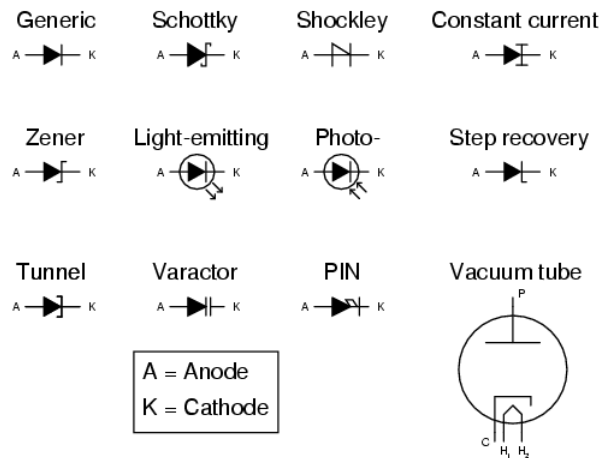
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9.10: Connectors



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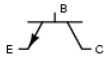
9.11: Diodes



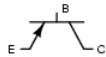
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9.12: Transistors, Bipolar

Bipolar NPN



Bipolar PNP



... with case

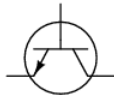
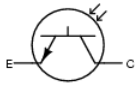
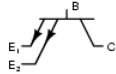


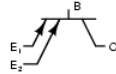
Photo-



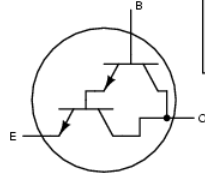
Dual-emitter NPN



Dual-emitter PNP

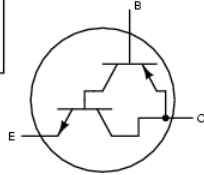


Darlington pair



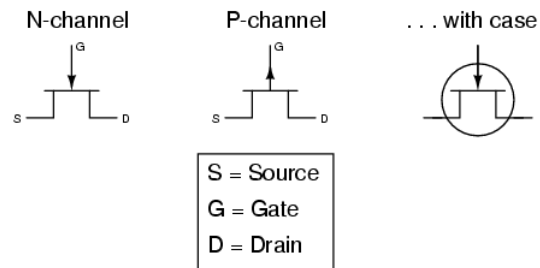
E = Emitter
B = Base
C = Collector

Sziklai pair



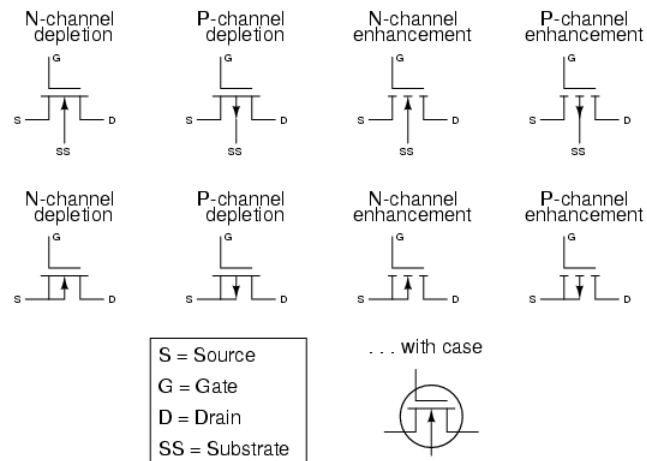
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9.13: Transistors, junction field-effect (JFET)



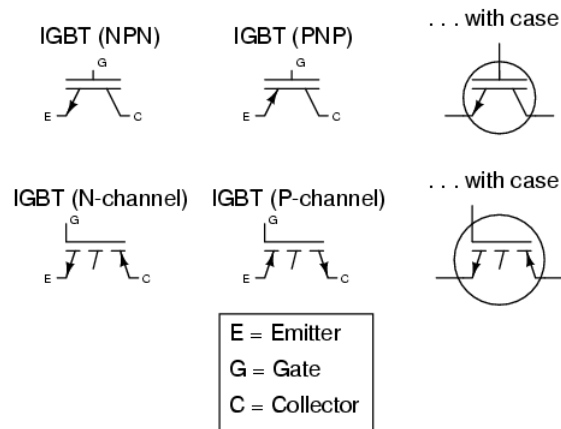
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9.14: Transistors, Insulated-gate Field-effect (IGFET or MOSFET)



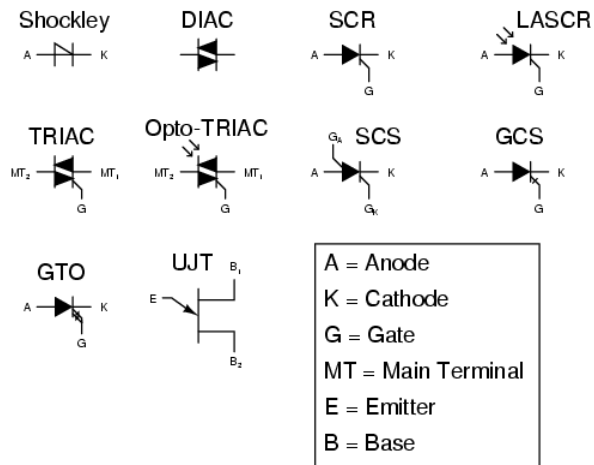
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9.15: Transistors, Hybrid



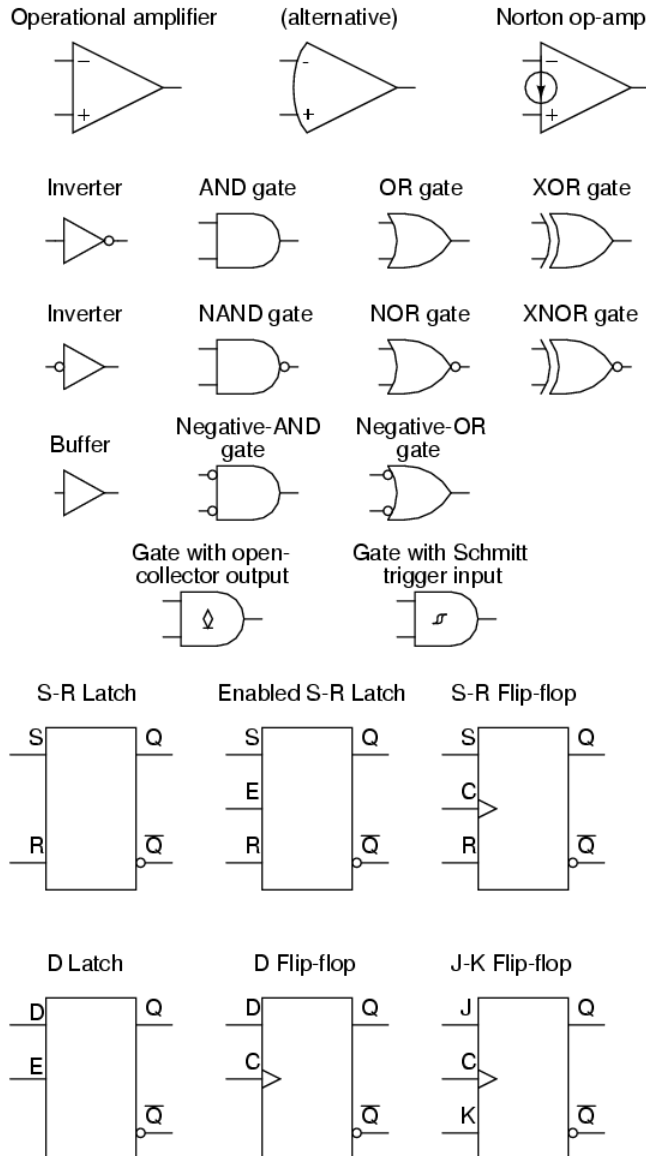
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9.16: The Thyristor



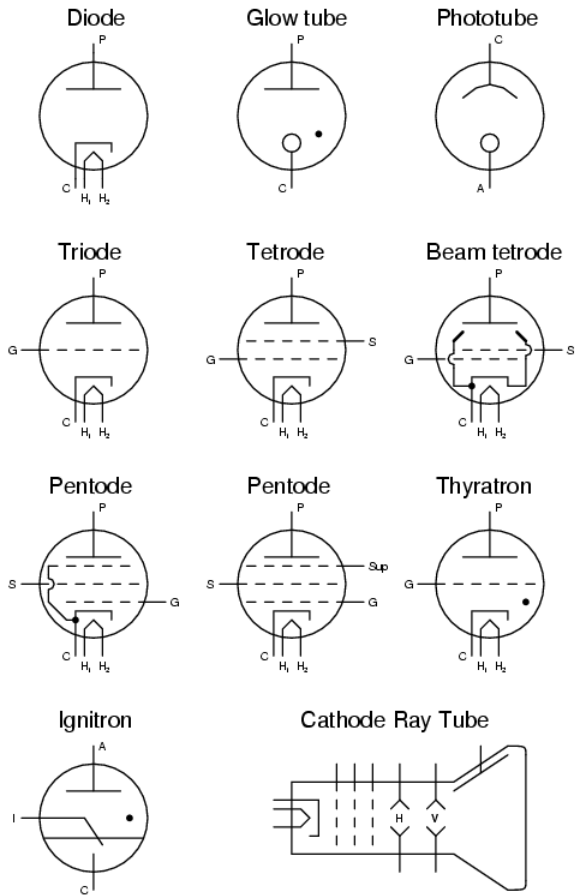
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9.17: Integrated Circuits



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9.18: Electron Tubes



P = Plate	S = Screen
G = Grid	A = Anode
C = Cathode	H = Heater
I = Ignitor	Sup = Suppressor

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CHAPTER OVERVIEW

10: Periodic Table Of The Elements

[10.1: Table \(landscape view\)](#)

[10.2: Table \(portrait view\)](#)

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10.1: Table (landscape view)

Periodic Table of the Elements

1 H Hydrogen 1.00794	2 He Helium 4.00260	3 Li Lithium 6.941	4 Be Beryllium 9.012182	5 B Boron 10.81	6 C Carbon 12.011	7 N Nitrogen 14.0067	8 O Oxygen 15.9994	9 F Fluorine 18.9984	10 Ne Neon 20.179	11 Na Sodium 22.989769	12 Mg Magnesium 24.3050	13 Al Aluminum 26.9815	14 Si Silicon 28.0855	15 P Phosphorus 30.9738	16 S Sulfur 32.06	17 Cl Chlorine 35.453	18 Ar Argon 39.948	19 K Potassium 39.0983	20 Ca Calcium 40.078	21 Sc Scandium 44.955910	22 Ti Titanium 47.88	23 V Vanadium 50.9415	24 Cr Chromium 51.9961	25 Mn Manganese 54.93805	26 Fe Iron 55.847	27 Co Cobalt 58.93320	28 Ni Nickel 58.69	29 Cu Copper 63.546	30 Zn Zinc 65.39	31 Ga Gallium 69.723	32 Ge Germanium 72.61	33 As Arsenic 74.92159	34 Se Selenium 78.96	35 Br Bromine 79.904	36 Kr Krypton 83.80	37 Rb Rubidium 85.4678	38 Sr Strontium 87.62	39 Y Yttrium 88.90585	40 Zr Zirconium 91.224	41 Nb Niobium 92.90638	42 Mo Molybdenum 95.94	43 Tc Technetium 98.90625	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.90550	46 Pd Palladium 106.42	47 Ag Silver 107.8682	48 Cd Cadmium 112.411	49 In Indium 114.82	50 Sn Tin 118.710	51 Sb Antimony 121.75	52 Te Tellurium 127.60	53 I Iodine 126.905	54 Xe Xenon 131.30	55 Cs Cesium 132.90543	56 Ba Barium 137.327	57-71 Lanthanide series	72 Hf Hafnium 178.49	73 Ta Tantalum 180.9479	74 W Tungsten 183.85	75 Re Rhenium 186.207	76 Os Osmium 190.2	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.96654	80 Hg Mercury 200.59	81 Tl Thallium 204.3833	82 Pb Lead 207.2	83 Bi Bismuth 208.98037	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)	87 Fr Francium (223)	88 Ra Radium (226)	89-103 Actinide series	104 Unq Ununquadium (261)	105 Uup Ununpentium (286)	106 Uuh Ununhexium (289)	107 Uus Ununseptium (304)	108 Uuo Ununoctium (315)	109 Uue Ununennium (318)	110 Ds Darmstadtium (310)	111 Nh Nihonium (315)	112 Fl Flerovium (318)	113 Nh Nihonium (315)	114 Fl Flerovium (318)	115 Mc Moscovium (315)	116 Lv Livermorium (310)	117 Ts Tennessine (315)	118 Og Oganesson (310)	119 Uut Ununtrium (310)	120 Uuq Ununquadium (315)	121 Uub Unubium (315)	122 Uud Unudunium (315)	123 Uus Ununseptium (315)	124 Uuo Ununoctium (315)	125 Uue Ununennium (315)	126 Uuh Ununhexium (315)	127 Uus Ununseptium (315)	128 Uuo Ununoctium (315)	129 Uue Ununennium (315)	130 Uuh Ununhexium (315)	131 Uus Ununseptium (315)	132 Uuo Ununoctium (315)	133 Uue Ununennium (315)	134 Uuh Ununhexium (315)	135 Uus Ununseptium (315)	136 Uuo Ununoctium (315)	137 Uue Ununennium (315)	138 Uuh Ununhexium (315)	139 Uus Ununseptium (315)	140 Uuo Ununoctium (315)	141 Uue Ununennium (315)	142 Uuh Ununhexium (315)	143 Uus Ununseptium (315)	144 Uuo Ununoctium (315)	145 Uue Ununennium (315)	146 Uuh Ununhexium (315)	147 Uus Ununseptium (315)	148 Uuo Ununoctium (315)	149 Uue Ununennium (315)	150 Uuh Ununhexium (315)	151 Uus Ununseptium (315)	152 Uuo Ununoctium (315)	153 Uue Ununennium (315)	154 Uuh Ununhexium (315)	155 Uus Ununseptium (315)	156 Uuo Ununoctium (315)	157 Uue Ununennium (315)	158 Uuh Ununhexium (315)	159 Uus Ununseptium 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Ununoctium (315)	437 Uue Ununennium (315)	438 Uuh Ununhexium (315)	439 Uus Ununseptium (315)	440 Uuo Ununoctium (315)	441 Uue Ununennium (315)	442 Uuh Ununhexium (315)	443 Uus Ununseptium (315)	444 Uuo Ununoctium (315)	445 Uue Ununennium (315)	446 Uuh Ununhexium (315)	447 Uus Ununseptium (315)	448 Uuo Ununoctium (315)	449 Uue Ununennium (315)	450 Uuh Ununhexium (315)	451 Uus Ununseptium (315)	452 Uuo Ununoctium (315)	453 Uue Ununennium (315)	454 Uuh Ununhexium (315)	455 Uus Ununseptium (315)	456 Uuo Ununoctium (315)	457 Uue Ununennium (315)	458 Uuh Ununhexium (315)	459 Uus Ununseptium (315)	460 Uuo Ununoctium (315)	461 Uue Ununennium (315)	462 Uuh Ununhexium (315)	463 Uus Ununseptium (315)	464 Uuo Ununoctium (315)	465 Uue Ununennium (315)	466 Uuh Ununhexium (315)	467 Uus Ununseptium (315)	468 Uuo Ununoctium (315)	469 Uue Ununennium (315)	470 Uuh Ununhexium (315)	471 Uus Ununseptium (315)	472 Uuo Ununoctium (315)	473 Uue Ununennium (315)	474 Uuh Ununhexium (315)	475 Uus Ununseptium (315)	476 Uuo Ununoctium (315)	477 Uue Ununennium (315)	478 Uuh Ununhexium (315)	479 Uus Ununseptium (315)	480 Uuo Ununoctium (315)	481 Uue Ununennium (315)	482 Uuh Ununhexium (315)	483 Uus Ununseptium (315)	484 Uuo Ununoctium (315)	485 Uue Ununennium (315)	486 Uuh Ununhexium (315)	487 Uus Ununseptium (315)	488 Uuo Ununoctium (315)	489 Uue Ununennium (315)	490 Uuh Ununhexium (315)	491 Uus Ununseptium (315)	492 Uuo Ununoctium (315)	493 Uue Ununennium (315)	494 Uuh Ununhexium (315)	495 Uus Ununseptium (315)	496 Uuo Ununoctium (315)	497 Uue Ununennium (315)	498 Uuh Ununhexium (315)	499 Uus Ununseptium (315)	500 Uuo Ununoctium (315)	501 Uue Ununennium (315)	502 Uuh Ununhexium (315)	503 Uus Ununseptium (315)	504 Uuo Ununoctium (315)	505 Uue Ununennium (315)	506 Uuh Ununhexium (315)	507 Uus Ununseptium (315)	508 Uuo Ununoctium (315)	509 Uue Ununennium (315)	510 Uuh Ununhexium (315)	511 Uus Ununseptium (315)	512 Uuo Ununoctium (315)	513 Uue Ununennium (315)	514 Uuh Ununhexium (315)	515 Uus Ununseptium (315)	516 Uuo Ununoctium (315)	517 Uue Ununennium (315)	518 Uuh Ununhexium (315)	519 Uus Ununseptium (315)	520 Uuo Ununoctium (315)	521 Uue Ununennium (315)	522 Uuh Ununhexium (315)	523 Uus Ununseptium (315)	524 Uuo Ununoctium (315)	525 Uue Ununennium (315)	526 Uuh Ununhexium (315)	527 Uus Ununseptium (315)	528 Uuo Ununoctium (315)	529 Uue Ununennium (315)	530 Uuh Ununhexium (315)	531 Uus Ununseptium (315)	532 Uuo Ununoctium (315)	533 Uue Ununennium (315)	534 Uuh Ununhexium (315)	535 Uus Ununseptium (315)	536 Uuo Ununoctium (315)	537 Uue Ununennium (315)	538 Uuh Ununhexium (315)	539 Uus Ununseptium (315)	540 Uuo Ununoctium (315)	541 Uue Ununennium (315)	542 Uuh Ununhexium (315)	543 Uus
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10.2: Table (portrait view)

Periodic Table of the Elements																		He 2 Helium 4.00260																	
1 IA H Hydrogen 1.00794																		18 VIIIA																	
2 IIA		Group new → 1 1A ← Group old																13 IIIA		14 IVA		15 VA		16 VIA		17 VIIA									
Symbol		Atomic number																B 5 Boron 10.81		C 6 Carbon 12.011		N 7 Nitrogen 14.007		O 8 Oxygen 15.9994		F 9 Fluorine 18.9984		Ne 10 Neon 20.179							
Name		Atomic mass is rounded according to convention on purity																																	
Electron configuration																		4s 1		3d 1		3p 1		3s 1		2p 1									
Metals																		Nonmetals																	
3 Li Lithium 6.941		4 Be Beryllium 9.012		5 B Boron 10.81		6 C Carbon 12.01		7 N Nitrogen 14.01		8 O Oxygen 16.00		9 F Fluorine 19.00		10 Ne Neon 20.18		11 Na Sodium 22.99		12 Mg Magnesium 24.31		13 Al Aluminum 26.98		14 Si Silicon 28.09		15 P Phosphorus 30.97		16 S Sulfur 32.06		17 Cl Chlorine 35.45		18 Ar Argon 36.96					
19 K Potassium 39.098		20 Ca Calcium 40.078		21 Sc Scandium 44.956		22 Ti Titanium 47.88		23 V Vanadium 50.942		24 Cr Chromium 51.996		25 Mn Manganese 54.938		26 Fe Iron 55.847		27 Co Cobalt 58.933		28 Ni Nickel 58.69		29 Cu Copper 63.546		30 Zn Zinc 65.39		31 Ga Gallium 69.723		32 Ge Germanium 72.61		33 As Arsenic 74.922		34 Se Selenium 78.96		35 Br Bromine 79.904		36 Kr Krypton 83.80	
37 Rb Rubidium 85.468		38 Sr Strontium 87.62		39 Y Yttrium 88.906		40 Zr Zirconium 91.224		41 Nb Niobium 92.906		42 Mo Molybdenum 95.94		43 Tc Technetium (98)		44 Ru Ruthenium (101)		45 Rh Rhodium (102)		46 Pd Palladium (106)		47 Ag Silver (108)		48 Cd Cadmium (112)		49 In Indium (114)		50 Sn Tin (119)		51 Sb Antimony (122)		52 Te Tellurium (128)		53 I Iodine (127)		54 Xe Xenon (131)	
55 Cs Cesium 132.905		56 Ba Barium 137.327		57-71 La-Lu Lanthanide series		72 Hf Hafnium (178)		73 Ta Tantalum (181)		74 W Tungsten (184)		75 Re Rhenium (186)		76 Os Osmium (190)		77 Ir Iridium (192)		78 Pt Platinum (195)		79 Au Gold (197)		80 Hg Mercury (201)		81 Tl Thallium (204)		82 Pb Lead (207)		83 Bi Bismuth (209)		84 Po Polonium (209)		85 At Astatine (210)		86 Rn Radon (222)	
87 Fr Francium (223)		88 Ra Radium (226)		89-103 Ac-Lr Actinide series		104 Unq Unquadium (261)		105 Unp Unpentium (262)		106 Unh Unhexium (263)		107 Uns Unseptium (264)		108 Uno Unoctium (265)		109 Unp Unpentium (266)		110 Unh Unhexium (267)		111 Uns Unseptium (268)		112 Uno Unoctium (269)		113 Unp Unpentium (270)		114 Unh Unhexium (271)		115 Uns Unseptium (272)		116 Uno Unoctium (273)		117 Unp Unpentium (274)		118 Unh Unhexium (275)	
119 Uu Ununennium (119)		120 Uu Unbinilium (120)		121-137 Uut-Uus Extended series		138 Uuh Unhennium (138)		139 Uub Unbibium (139)		140 Uuq Unquadium (140)		141 Uup Unpentium (141)		142 Uuh Unhennium (142)		143 Uub Unbibium (143)		144 Uuq Unquadium (144)		145 Uup Unpentium (145)		146 Uuh Unhennium (146)		147 Uub Unbibium (147)		148 Uuq Unquadium (148)		149 Uup Unpentium (149)		150 Uuh Unhennium (150)		151 Uub Unbibium (151)		152 Uuq Unquadium (152)	
153 Uut Untrium (153)		154 Uuq Unquadium (154)		155 Uup Unpentium (155)		156 Uuh Unhennium (156)		157 Uub Unbibium (157)		158 Uuq Unquadium (158)		159 Uup Unpentium (159)		160 Uuh Unhennium (160)		161 Uub Unbibium (161)		162 Uuq Unquadium (162)		163 Uup Unpentium (163)		164 Uuh Unhennium (164)		165 Uub Unbibium (165)		166 Uuq Unquadium (166)		167 Uup Unpentium (167)		168 Uuh Unhennium (168)		169 Uub Unbibium (169)		170 Uuq Unquadium (170)	
171 Uus Unseptium (171)		172 Uuh Unhennium (172)		173 Uub Unbibium (173)		174 Uuq Unquadium (174)		175 Uup Unpentium (175)		176 Uuh Unhennium (176)		177 Uub Unbibium (177)		178 Uuq Unquadium (178)		179 Uup Unpentium (179)		180 Uuh Unhennium (180)		181 Uub Unbibium (181)		182 Uuq Unquadium (182)		183 Uup Unpentium (183)		184 Uuh Unhennium (184)		185 Uub Unbibium (185)		186 Uuq Unquadium (186)		187 Uup Unpentium (187)		188 Uuh Unhennium (188)	
189 Uus Unseptium (189)		190 Uuh Unhennium (190)		191 Uub Unbibium (191)		192 Uuq Unquadium (192)		193 Uup Unpentium (193)		194 Uuh Unhennium (194)		195 Uub Unbibium (195)		196 Uuq Unquadium (196)		197 Uup Unpentium (197)		198 Uuh Unhennium (198)		199 Uub Unbibium (199)		200 Uuq Unquadium (200)		201 Uup Unpentium (201)		202 Uuh Unhennium (202)		203 Uub Unbibium (203)		204 Uuq Unquadium (204)		205 Uup Unpentium (205)		206 Uuh Unhennium (206)	
207 Uus Unseptium (207)		208 Uuh Unhennium (208)		209 Uub Unbibium (209)		210 Uuq Unquadium (210)		211 Uup Unpentium (211)		212 Uuh Unhennium (212)		213 Uub Unbibium (213)		214 Uuq Unquadium (214)		215 Uup Unpentium (215)		216 Uuh Unhennium (216)		217 Uub Unbibium (217)		218 Uuq Unquadium (218)		219 Uup Unpentium (219)		220 Uuh Unhennium (220)		221 Uub Unbibium (221)		222 Uuq Unquadium (222)		223 Uup Unpentium (223)		224 Uuh Unhennium (224)	
225 Uus Unseptium (225)		226 Uuh Unhennium (226)		227 Uub Unbibium (227)		228 Uuq Unquadium (228)		229 Uup Unpentium (229)		230 Uuh Unhennium (230)		231 Uub Unbibium (231)		232 Uuq Unquadium (232)		233 Uup Unpentium (233)		234 Uuh Unhennium (234)		235 Uub Unbibium (235)		236 Uuq Unquadium (236)		237 Uup Unpentium (237)		238 Uuh Unhennium (238)		239 Uub Unbibium (239)		240 Uuq Unquadium (240)		241 Uup Unpentium (241)		242 Uuh Unhennium (242)	
243 Uus Unseptium (243)		244 Uuh Unhennium (244)		245 Uub Unbibium (245)		246 Uuq Unquadium (246)		247 Uup Unpentium (247)		248 Uuh Unhennium (248)		249 Uub Unbibium (249)		250 Uuq Unquadium (250)		251 Uup Unpentium (251)		252 Uuh Unhennium (252)		253 Uub Unbibium (253)		254 Uuq Unquadium (254)		255 Uup Unpentium (255)		256 Uuh Unhennium (256)		257 Uub Unbibium (257)		258 Uuq Unquadium (258)		259 Uup Unpentium (259)		260 Uuh Unhennium (260)	
261 Uus Unseptium (261)		262 Uuh Unhennium (262)		263 Uub Unbibium (263)		264 Uuq Unquadium (264)		265 Uup Unpentium (265)		266 Uuh Unhennium (266)		267 Uub Unbibium (267)		268 Uuq Unquadium (268)		269 Uup Unpentium (269)		270 Uuh Unhennium (270)		271 Uub Unbibium (271)		272 Uuq Unquadium (272)		273 Uup Unpentium (273)		274 Uuh Unhennium (274)		275 Uub Unbibium (275)		276 Uuq Unquadium (276)		277 Uup Unpentium (277)		278 Uuh Unhennium (278)	
279 Uus Unseptium (279)		280 Uuh Unhennium (280)		281 Uub Unbibium (281)		282 Uuq Unquadium (282)		283 Uup Unpentium (283)		284 Uuh Unhennium (284)		285 Uub Unbibium (285)		286 Uuq Unquadium (286)		287 Uup Unpentium (287)		288 Uuh Unhennium (288)		289 Uub Unbibium (289)		290 Uuq Unquadium (290)		291 Uup Unpentium (291)		292 Uuh Unhennium (292)		293 Uub Unbibium (293)		294 Uuq Unquadium (294)		295 Uup Unpentium (295)		296 Uuh Unhennium (296)	
297 Uus Unseptium (297)		298 Uuh Unhennium (298)		299 Uub Unbibium (299)		300 Uuq Unquadium (300)		301 Uup Unpentium (301)		302 Uuh Unhennium (302)		303 Uub Unbibium (303)		304 Uuq Unquadium (304)		305 Uup Unpentium (305)		306 Uuh Unhennium (306)		307 Uub Unbibium (307)		308 Uuq Unquadium (308)		309 Uup Unpentium (309)		310 Uuh Unhennium (310)		311 Uub Unbibium (311)		312 Uuq Unquadium (312)		313 Uup Unpentium (313)		314 Uuh Unhennium (314)	
315 Uus Unseptium (315)		316 Uuh Unhennium (316)		317 Uub Unbibium (317)		318 Uuq Unquadium (318)		319 Uup Unpentium (319)		320 Uuh Unhennium (320)		321 Uub Unbibium (321)		322 Uuq Unquadium (322)		323 Uup Unpentium (323)		324 Uuh Unhennium (324)		325 Uub Unbibium (325)		326 Uuq Unquadium (326)		327 Uup Unpentium (327)		328 Uuh Unhennium (328)		329 Uub Unbibium (329)		330 Uuq Unquadium (330)		331 Uup Unpentium (331)		332 Uuh Unhennium (332)	
333 Uus Unseptium (333)		334 Uuh Unhennium (334)		335 Uub Unbibium (335)		336 Uuq Unquadium (336)		337 Uup Unpentium (337)		338 Uuh Unhennium (338)		339 Uub Unbibium (339)		340 Uuq Unquadium (340)		341 Uup Unpentium (341)		342 Uuh Unhennium (342)		343 Uub Unbibium (343)		344 Uuq Unquadium (344)		345 Uup Unpentium (345)		346 Uuh Unhennium (346)		347 Uub Unbibium (347)		348 Uuq Unquadium (348)		349 Uup Unpentium (349)		350 Uuh Unhennium (350)	
351 Uus Unseptium (351)		352 Uuh Unhennium (352)		353 Uub Unbibium (353)		354 Uuq Unquadium (354)		355 Uup Unpentium (355)		356 Uuh Unhennium (356)		357 Uub Unbibium (357)		358 Uuq Unquadium (358)		359 Uup Unpentium (359)		360 Uuh Unhennium (360)		361 Uub Unbibium (361)		362 Uuq Unquadium (362)		363 Uup Unpentium (363)		364 Uuh Unhennium (364)		365 Uub Unbibium (365)		366 Uuq Unquadium (366)		367 Uup Unpentium (367)		368 Uuh Unhennium (368)	
369 Uus Unseptium (369)		370 Uuh Unhennium (370)		371 Uub Unbibium (371)		372 Uuq Unquadium (372)		373 Uup Unpentium (373)		374 Uuh Unhennium (374)		375 Uub Unbibium (375)		376 Uuq Unquadium (376)		377 Uup Unpentium (377)		378 Uuh Unhennium (378)		379 Uub Unbibium (379)		380 Uuq Unquadium (380)		381 Uup Unpentium (381)		382 Uuh Unhennium (382)		383 Uub Unbibium (383)		384 Uuq Unquadium (384)		385 Uup Unpentium (385)		386 Uuh Unhennium (386)	
387 Uus Unseptium (387)		388 Uuh Unhennium (388)		389 Uub Unbibium (389)		390 Uuq Unquadium (390)		391 Uup Unpentium (391)		392 Uuh Unhennium (392)		393 Uub Unbibium (393)		394 Uuq Unquadium (394)		395 Uup Unpentium (395)		396 Uuh Unhennium (396)		397 Uub Unbibium (397)		398 Uuq Unquadium (398)		399 Uup Unpentium (399)		400 Uuh Unhennium (400)		401 Uub Unbibium (401)		402 Uuq Unquadium (402)		403 Uup Unpentium (403)		404 Uuh Unhennium (404)	
405 Uus Unseptium (405)		406 Uuh Unhennium (406)		407 Uub Unbibium (407)		408 Uuq Unquadium (408)		409 Uup Unpentium (409)		410 Uuh Unhennium (410)		411 Uub Unbibium (411)		412 Uuq Unquadium (412)		413 Uup Unpentium (413)		414 Uuh Unhennium (414)		415 Uub Unbibium (415)		416 Uuq Unquadium (416)		417 Uup Unpentium (417)		418 Uuh Unhennium (418)		419 Uub Unbibium (419)		420 Uuq Unquadium (420)		421 Uup Unpentium (421)		422 Uuh Unhennium (422)	
423 Uus Unseptium (423)		424 Uuh Unhennium (424)		425 Uub Unbibium (425)		426 Uuq Unquadium (426)		427 Uup Unpentium (427)		428 Uuh Unhennium (428)		429 Uub Unbibium (429)		430 Uuq Unquadium (430)		431 Uup Unpentium (431)		432 Uuh Unhennium (432)		433 Uub Unbibium (433)		434 Uuq Unquadium (434)		435 Uup Unpentium (435)		436 Uuh Unhennium (436)		437 Uub Unbibium (437)		438 Uuq Unquadium (438)		439 Uup Unpentium (439)		440 Uuh Unhennium (440)	
441 Uus Unseptium (441)		442 Uuh Unhennium (442)		443 Uub Unbibium (443)		444 Uuq Unquadium (444)		445 Uup Unpentium (445)		446 Uuh Unhennium (446)		447 Uub Unbibium (447)		448 Uuq Unquadium (448)		449 Uup Unpentium (449)		450 Uuh Unhennium (450)		451 Uub Unbibium (451)		452 Uuq Unquadium (452)		453 Uup Unpentium (453)		454 Uuh Unhennium (454)		455 Uub Unbibium (455)		456 Uuq Unquadium (456)		457 Uup Unpentium (457)		458 Uuh Unhennium (458)	
459 Uus Unseptium (459)		460 Uuh Unhennium (460)		461 Uub Unbibium (461)		462 Uuq Unquadium (462)		463 Uup Unpentium (463)		464 Uuh Unhennium (464)		465 Uub Unbibium (465)		466 Uuq Unquadium (466)		467 Uup Unpentium (467)		468 Uuh Unhennium (468)		469 Uub Unbibium (469)		470 Uuq Unquadium (470)		471 Uup Unpentium (471)		472 Uuh Unhennium (472)		473 Uub Unbibium (473)		474 Uuq Unquadium (474)		475 Uup Unpentium (475)		476 Uuh Unhennium (476)	
477 Uus Unseptium (477)		478 Uuh Unhennium (478)		479 Uub Unbibium (479)		480 Uuq Unquadium (480)		481 Uup Unpentium (481)		482 Uuh Unhennium (482)		483 Uub Unbibium (483)		484 Uuq Unquadium (484)		485 Uup Unpentium (485)		486 Uuh Unhennium (486)		487 Uub Unbibium (487)		488 Uuq Unquadium (488)		489 Uup Unpentium (489)		490 Uuh Unhennium (490)		491 Uub Unbibium (491)		492 Uuq Unquadium (492)		493 Uup Unpentium (493)		494 Uuh Unhennium (494)	
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Periodic table of chemical elements.

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All entries arranged in alphabetical order of surname. Major contributions are listed by individual name with some detail on the nature of the contribution(s), date, contact info, etc. Minor contributions (typo corrections, etc.) are listed by name only for reasons of brevity. Please understand that when I classify a contribution as “minor,” it is in no way inferior to the effort or value of a “major” contribution, just smaller in the sense of less text changed. Any and all contributions are gratefully accepted. I am indebted to all those who have given freely of their own knowledge, time, and resources to make this a better book!

Dennis Crunkilton

- **Date(s) of contribution(s):** October 2005 to present
- **Nature of contribution:** Ch 1, added permittivity, capacitor and inductor formulas, wire table; 10/2005.
- **Nature of contribution:** Ch 1, expanded dielectric table, 10232.eps, copied data from Volume 1, Chapter 13; 10/2005.
- **Nature of contribution:** Mini table of contents, all chapters except appendices; html, latex, ps, pdf; See Devel/tutorial.html; 01/2006.
- **Nature of contribution:** Changed CH2 from “Resistor color codes” to “Color codes”; Added wiring color codes; 10/2007.
- **Contact at:** dcrunkilton(at)att(dot)net

Alejandro Gamero Divasto

- **Date(s) of contribution(s):** January 2002
- **Nature of contribution:** Suggestions related to troubleshooting: caveat regarding swapping two similar components as a troubleshooting tool; avoiding pressure placed on the troubleshooter; perils of “team” troubleshooting; wisdom of recording system history; operator error as a cause of failure; and the perils of finger-pointing.

Tony R. Kuphaldt

- **Date(s) of contribution(s):** 1996 to present
- **Nature of contribution:** Original author.
- **Contact at:** liec0@lycos.com

Your name here

- **Date(s) of contribution(s):** Month and year of contribution
- **Nature of contribution:** Insert text here, describing how you contributed to the book.
- **Contact at:** my_email@provider.net

Typo corrections and other “minor” contributions

- *The students of Bellingham Technical College’s Instrumentation program.*
- **Bernard Sheehan** (January 2005), Typographical error correction in “Right triangle trigonometry” section Chapter 5: TRIGONOMETRY REFERENCE (two formulas for $\tan x$ the second one reads $\tan x = \cos x / \sin x$ it should be $\cot x = \cos x / \sin x$ — changes to 01001.eps previously made)
- **Michiel van Bolhuis** (April 2007) Typo Ch 1, s/picofards/picofarads.
- **Chirvasuta Constantin** (April 2003) Identified error in quadratic equation formula.
- **Colin Creitz** (May 2007) Chapters: several, s/it’s/its.
- **Jeff DeFreitas** (March 2006) Improve appearance: replace “/” and ”/” Chapters: A1, A2.
- **Gerald Gardner** (January 2003) Suggested adding Imperial gallons conversion to table.
- **Geoff Hosking** (July 2006) Typo correction in Conductors and Insulators chapter, Critical Temperatures of Superconductors: s/degrees Kelvin/Kelvins.
- **Harvey Lew** (??? 2003) Typo correction in Trig chapter: “tangent” should have been “cotangent”.
- **Len Nunn** (May 2008) Typo correction in Calculus chapter: “ $dx/d(a^x)$ ” in error, 11042.png .
- **Don Stalkowski** (June 2002) Technical help with PostScript-to-PDF file format conversion.
- **Joseph Teichman** (June 2002) Suggestion and technical help regarding use of PNG images instead of JPEG.
- **Mark44@allaboutcircuits.com** (March 2008) Ch 4, Clarification of division by zero.
- **Timothy Unregistered@allaboutcircuits.com** (Feb 2008) Changed default roman font to newcent.

- **Imranullah Syed** (Feb 2008) Suggested centering of uncaptioned schematics.
- Unregistered@allaboutcircuits.com (Aug 2008) formatting of PDF off pps 130-136.
- **D Crunkilton** (Dec 2009) added missing images 10232.eps 10233.eps 10238.eps 10239.eps 10241.eps
- webbie@allaboutcircuits.com (Aug 2010) Ch 1, s/usefull/useful/.
- **D. Crunkilton** (June 2011) hi.latex, header file; updated link to openbookproject.net .

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